

Chapter 1

Introduction

1.1 Introduction

Modern high-energy physics seeks to understand the elementary constituents of matter and the fundamental interactions that govern their behaviour. At present, the most successful theoretical description of particle physics is the Standard Model (SM), a quantum field theory based on the gauge symmetry

$$SU(3)_C \times SU(2)_L \times U(1)_Y. \quad (1.1)$$

Within this framework, matter is described by three generations of quarks and leptons, while the electromagnetic, weak, and strong interactions are mediated by gauge bosons. The electroweak symmetry is spontaneously broken through the Higgs mechanism, giving masses to the weak gauge bosons and to the fermions through their interactions with the Higgs field. The discovery of the Higgs boson by the ATLAS and CMS experiments in 2012 completed the particle content predicted by the SM and marked a major milestone in modern particle physics.

The SM has been tested with remarkable precision over a wide range of energy scales and has successfully described a vast number of experimental observations. Nevertheless, it is widely regarded as an incomplete theory. It does not provide a particle candidate for dark matter, does not explain the observed matter–antimatter asymmetry of the universe, and does not account for the origin of non-zero neutrino masses. In addition, the pattern of fermion masses and flavour mixing remains unexplained, and the large separation between the electroweak and Planck scales suggests the possible existence of new physics beyond the SM. These open questions motivate direct searches for new particles and interactions at the highest experimentally accessible energies.

The Large Hadron Collider (LHC) at CERN provides a unique environment for probing such new phenomena. By colliding protons at tera-electronvolt centre-of-mass energies, the LHC allows the production of particles that may be associated with physics beyond the SM. During Run 2 of the LHC from year 2015 to 2018, proton–proton (pp) collisions were delivered at a centre-of-mass energy of $\sqrt{s} = 13$ TeV.

ATLAS is a detector located at one of the interaction points of the LHC. It is designed to observe and study a wide range of physics processes, mainly from the pp collisions, from precision measurements of SM interactions to searches for new particles. The detector consists of several sub-detectors surrounding the collision interaction point, including an inner tracking detector, electromagnetic and hadronic calorimeters, and a muon spectrometer. Together with a powerful magnet system, these components allow the reconstruction and identification of electrons, photons, muons, hadronically decaying τ -leptons, jets, and missing transverse momentum, denoted by E_T^{miss} .

Among the many possible extensions of the SM, leptoquarks (LQs) provide a well-motivated class of new particles connecting the quark and lepton sectors. They are hypothetical bosons that couple directly to a quark and a lepton, and they appear naturally in theories such as grand unified theories and models of quark–lepton unification. Leptoquarks are also of particular phenomenological interest because they can modify semileptonic processes and may therefore be connected to the observed tensions in flavour observables.

A search for the leptoquarks is presented in the thesis using 140 fb^{-1} of pp collision data recorded by the ATLAS detector at a centre-of-mass energy of $\sqrt{s} = 13$ TeV. This thesis focuses on the vector singlet leptoquark U_1 as the benchmark model. This model can contribute to processes involving b quarks, τ leptons, and neutrinos. Therefore, the analysis targets events with a hadronically decaying τ -lepton, large missing transverse momentum, and jets. The observed data are compared with SM background predictions in signal regions designed to enhance sensitivity to possible leptoquark contributions. In the absence of a significant excess over the SM expectation, constraints are placed on the allowed parameter space of the U_1 model, expressed in terms of the leptoquark mass and its couplings.

This dissertation is organised as follows. Chapter 2 provides an overview of the SM’s theoretical framework and the physics beyond it. Chapter 3 describes the leptoquark models and the searches performed at LHC. Chapter 4 introduces the LHC and the ATLAS detector. Chapter 5 provides the dataset and simulated samples. Chapter 6 details the reconstruction and the identification of the physics objects that are used for event selection. Chapter 7 discusses the optimisation for the event selection used for signal signature extraction. Chapter 8 presents the background estimation strategy. Chapter 9 discusses the systematic

uncertainties for the background and signals. Chapter 10 presents the statistical interpretation of the results. Chapter 11 summarises this analysis.

Chapter 2

Theoretical Background

This chapter introduces the theoretical background which motivates this search performed in the thesis. The discussion begins with the Standard Model (SM), which provides the current theoretical framework for describing elementary particles and their interactions. The limitations of the SM are then summarised, followed by a discussion of several possible extensions beyond the SM. Particular emphasis is placed on leptoquark models, which provide a direct connection between the quark and lepton sectors and motivate the experimental search presented in this thesis.

2.1 The Standard Model

The Standard Model (SM) is a relativistic quantum field theory describing the known elementary particles and their strong, weak, and electromagnetic interactions. Its structure is determined by local gauge invariance under

$$SU(3)_C \times SU(2)_L \times U(1)_Y. \quad (2.1)$$

The group $SU(3)_C$ describes the colour symmetry of the strong interaction, while $SU(2)_L \times U(1)_Y$ describes the electroweak interaction [1–3]. In quantum field theory, particles are classified not only by their mass and spin, but also by their transformation properties under these gauge symmetries. Particles with half-integer spin are fermions and obey Fermi–Dirac statistics, while particles with integer spin are bosons and obey Bose–Einstein statistics [4]. The matter particles of the SM are spin-1/2 fermions, the interaction carriers are spin-1 gauge bosons, and the Higgs boson is a spin-0 scalar boson.

The minimal SM Lagrangian used here is constructed from the SM field content and is invariant under the local gauge group. Therefore, instead of introducing the Lagrangian as

an isolated expression, this section first summarises the field content and gauge quantum numbers of the SM. The principle of local gauge invariance is then introduced, followed by the gauge transformations of the strong sector, based on $SU(3)_C$, and of the electroweak sector, based on $SU(2)_L \times U(1)_Y$. The covariant derivative and field-strength tensors are then constructed from these symmetries, leading to the full SM Lagrangian. Finally, electroweak symmetry breaking and the generation of fermion masses are discussed.

2.1.1 Particle content and gauge group of the SM

The gauge quantum numbers of a field specify how it transforms under the SM gauge group. In this thesis, the electric charge is written as

$$Q = T^3 + Y, \quad (2.2)$$

where T^3 is the third component of weak isospin and Y is the weak hypercharge. This convention is used throughout the thesis.

For each generation, the SM fermions are arranged as

$$Q_L^i = \begin{pmatrix} u_L^i \\ d_L^i \end{pmatrix} \sim (3, 2, 1/6), \quad u_R^i \sim (3, 1, 2/3), \quad d_R^i \sim (3, 1, -1/3), \quad (2.3)$$

$$L_L^i = \begin{pmatrix} \nu_L^i \\ e_L^i \end{pmatrix} \sim (1, 2, -1/2), \quad e_R^i \sim (1, 1, -1), \quad (2.4)$$

where $i = 1, 2, 3$ is the generation index, and the notation denotes the representation under $SU(3)_C \times SU(2)_L \times U(1)_Y$, which are the QCD representation, the weak isospin representation and the weak hypercharge, respectively. In the minimal SM, no right-handed neutrino field is included. The scalar sector contains one complex Higgs doublet,

$$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} \sim (1, 2, 1/2). \quad (2.5)$$

The gauge bosons are associated with the generators of the three gauge groups. The eight gluon fields G_μ^a , with $a = 1, \dots, 8$, are associated with $SU(3)_C$. The three weak gauge fields W_μ^I , with $I = 1, 2, 3$, are associated with $SU(2)_L$. The hypercharge gauge field B_μ is associated with $U(1)_Y$. Before electroweak symmetry breaking, W_μ^1 , W_μ^2 , W_μ^3 , and B_μ are gauge eigenstates. After electroweak symmetry breaking, they are reorganised into the physical electroweak bosons $W^\pm$, Z , and γ .

The main particle content of the SM is summarised in Table 2.1. For fermions and the Higgs boson, the table lists the corresponding electroweak quantum numbers using the convention in Eq. (2.2). For gauge bosons, the entries describe their gauge fields and the electric charges of the physical states where appropriate.

Table 2.1 Particle content of the SM and the corresponding spin and electroweak quantum numbers. For fermions and the Higgs doublet, T^3 and Y denote the third component of weak isospin and the weak hypercharge, respectively, with the electric charge defined as $Q = T^3 + Y$. For gauge bosons, only the electric charges of the physical states after electroweak symmetry breaking are shown.

Category	Type	Generation			Spin	Quantum numbers		
		1st	2nd	3rd		T^3	Y	Q
Fermion	Left-handed quark	u_L	c_L	t_L	1/2	+1/2	1/6	+2/3
		d_L	s_L	b_L		-1/2		-1/3
	Right-handed up-type quark	u_R	c_R	t_R	1/2	0	2/3	+2/3
	Right-handed down-type quark	d_R	s_R	b_R	1/2	0	-1/3	-1/3
	Left-handed lepton	ν_{eL}	$\nu_{\mu L}$	$\nu_{\tau L}$	1/2	+1/2	-1/2	0
		e_L	μ_L	τ_L		-1/2		-1
	Right-handed charged lepton	e_R	μ_R	τ_R	1/2	0	-1	-1
Gauge boson	Gluon	$g^a, a = 1, \dots, 8$			1	-	-	0
	Photon	γ			1	-	-	0
	Charged weak boson	$W^\pm$			1	-	-	± 1
	Neutral weak boson	Z			1	-	-	0
Scalar field	Higgs doublet	ϕ^+			0	+1/2	1/2	+1
		ϕ^0				-1/2		0

The quantum numbers listed above determine which interactions each field can have. They also determine which terms can be written in the SM Lagrangian. For example, explicit Dirac mass terms for charged fermions are not gauge invariant before electroweak symmetry breaking, because the left- and right-handed fermions transform differently under $SU(2)_L \times U(1)_Y$. Fermion masses instead arise from gauge-invariant Yukawa interactions with the Higgs field.

2.1.2 Local gauge invariance

The guiding principle in constructing the SM is local gauge invariance. A global transformation acts in the same way at every point in spacetime, while a local gauge transformation depends on the spacetime coordinate x . For a field $\psi(x)$ transforming under a gauge group with generators T^A , a generic local transformation is

$$\psi(x) \rightarrow \psi'(x) = U(x)\psi(x), \quad U(x) = \exp \left[i\alpha^A(x)T^A \right]. \quad (2.6)$$

Here $\alpha^A(x)$ are arbitrary real, spacetime-dependent gauge parameters. They specify how the field is transformed locally along each generator T^A of the gauge group, and the repeated index A is summed over all generators.

The ordinary derivative of the transformed field is

$$\partial_\mu \psi'(x) = U(x) \partial_\mu \psi(x) + (\partial_\mu U(x)) \psi(x). \quad (2.7)$$

The second term in Eq. (2.7) shows that $\partial_\mu \psi$ does not transform in the same way as ψ . Therefore, the kinetic term constructed using the ordinary derivative is not invariant under a local gauge transformation.

To restore local gauge invariance, the ordinary derivative is promoted to a covariant derivative D_μ satisfying

$$D_\mu \psi(x) \rightarrow D'_\mu \psi'(x) = U(x) D_\mu \psi(x). \quad (2.8)$$

This requirement introduces gauge fields. The gauge fields are therefore not added by hand as arbitrary new particles; they are required in order to compare fields at neighbouring spacetime points in a way that is independent of the local gauge convention. The following subsections describe how this principle is realised for the strong and electroweak gauge sectors of the SM.

2.1.3 The $SU(3)_C$ gauge sector

The strong interaction is described by quantum chromodynamics (QCD), a gauge theory based on the colour group $SU(3)_C$. Quark fields transform as colour triplets, while leptons and the Higgs field are colour singlets. For a colour triplet quark field $q(x)$, the local $SU(3)_C$ transformation is

$$q(x) \rightarrow q'(x) = U_C(x) q(x), \quad U_C(x) = \exp \left[i \alpha_C^a(x) \frac{\lambda^a}{2} \right], \quad (2.9)$$

where λ^a are the Gell-Mann matrices and $a = 1, \dots, 8$. The corresponding gauge fields are the eight gluon fields G_μ^a .

It is useful to introduce the matrix-valued gluon field

$$G_\mu(x) = G_\mu^a(x) \frac{\lambda^a}{2}. \quad (2.10)$$

With the convention

$$D_\mu = \partial_\mu - ig_s G_\mu^a \frac{\lambda^a}{2}, \quad (2.11)$$

where g_s is the strong coupling constant, the gluon field transforms as

$$G_\mu(x) \rightarrow G'_\mu(x) = U_C(x) G_\mu(x) U_C^\dagger(x) - \frac{i}{g_s} (\partial_\mu U_C(x)) U_C^\dagger(x). \quad (2.12)$$

This transformation law ensures that $D_\mu q$ transforms in the same way as q under local $SU(3)_C$ transformations.

The $SU(3)_C$ field-strength tensor is

$$G_{\mu\nu}^a = \partial_\mu G_\nu^a - \partial_\nu G_\mu^a + g_s f^{abc} G_\mu^b G_\nu^c, \quad (2.13)$$

where f^{abc} are the structure constants of $SU(3)$, defined by the commutation relation

$$\left[\frac{\lambda^a}{2}, \frac{\lambda^b}{2} \right] = i f^{abc} \frac{\lambda^c}{2}. \quad (2.14)$$

The last term in Eq. (2.13) arises from the non-Abelian nature of $SU(3)_C$. It gives rise to gluon self-interactions, which are a defining feature of QCD and are essential for asymptotic freedom at high energies and confinement at low energies.

2.1.4 The electroweak gauge sector

Before electroweak symmetry breaking, the weak and electromagnetic interactions are described together by the gauge group $SU(2)_L \times U(1)_Y$ [1–3]. The $SU(2)_L$ factor is the non-Abelian weak-isospin gauge symmetry, while the $U(1)_Y$ factor is the Abelian hypercharge gauge symmetry. They are introduced separately below, but the physical photon and Z boson arise only after electroweak symmetry breaking, when the neutral fields W_μ^3 and B_μ mix.

The $SU(2)_L$ gauge transformation

The $SU(2)_L$ gauge symmetry acts on left-handed fermion doublets and on the Higgs doublet:

$$Q_L^i, \quad L_L^i, \quad \Phi. \quad (2.15)$$

For an $SU(2)_L$ doublet $\chi(x)$, the local transformation is

$$\chi(x) \rightarrow \chi'(x) = U_L(x)\chi(x), \quad U_L(x) = \exp \left[i\alpha^I(x) \frac{\tau^I}{2} \right], \quad (2.16)$$

where τ^I are the Pauli matrices and $I = 1, 2, 3$. The right-handed charged fermions u_R^i, d_R^i , and e_R^i are singlets under $SU(2)_L$ and therefore have $T_L^I = 0$.

The three gauge fields associated with $SU(2)_L$ are denoted by W_μ^I . They can be written in matrix form as

$$W_\mu(x) = W_\mu^I(x) \frac{\tau^I}{2}. \quad (2.17)$$

With the convention

$$D_\mu = \partial_\mu - igW_\mu^I \frac{\tau^I}{2}, \quad (2.18)$$

where g is the $SU(2)_L$ coupling constant, the weak gauge field transforms as

$$W_\mu(x) \rightarrow W'_\mu(x) = U_L(x)W_\mu(x)U_L^\dagger(x) - \frac{i}{g} (\partial_\mu U_L(x)) U_L^\dagger(x). \quad (2.19)$$

This transformation law is the non-Abelian generalisation of the Abelian gauge-field transformation. The matrix structure reflects the fact that the $SU(2)_L$ generators do not commute.

The $SU(2)_L$ field-strength tensor is

$$W_{\mu\nu}^I = \partial_\mu W_\nu^I - \partial_\nu W_\mu^I + g\epsilon^{IJK} W_\mu^J W_\nu^K. \quad (2.20)$$

The last term arises from the non-Abelian nature of $SU(2)_L$ and leads to self-interactions among weak gauge bosons.

The $U(1)_Y$ gauge transformation

The $U(1)_Y$ gauge symmetry is the Abelian part of the electroweak gauge group. Its corresponding gauge charge is the weak hypercharge Y , which determines how each field transforms under a local $U(1)_Y$ phase rotation. For a field $\psi(x)$ with hypercharge Y , the local $U(1)_Y$ transformation is

$$\psi(x) \rightarrow \psi'(x) = \exp[iY\beta(x)] \psi(x), \quad (2.21)$$

where $\beta(x)$ is a spacetime-dependent transformation parameter. The corresponding gauge field is the hypercharge field B_μ . With the convention

$$D_\mu = \partial_\mu - ig'YB_\mu, \quad (2.22)$$

the gauge field transforms as

$$B_\mu(x) \rightarrow B'_\mu(x) = B_\mu(x) + \frac{1}{g'}\partial_\mu\beta(x). \quad (2.23)$$

This transformation ensures that $D_\mu\psi$ transforms in the same way as ψ .

The hypercharge assignments in Table 2.1 are chosen so that the electric charge is obtained from Eq. (2.2). For the left-handed lepton doublet, $Y = -1/2$. The upper component has $T^3 = +1/2$ and therefore $Q = 0$, corresponding to the neutrino. The lower component has $T^3 = -1/2$ and therefore $Q = -1$, corresponding to the charged lepton. Similarly, the left-handed quark doublet has $Y = 1/6$, giving charges $+2/3$ and $-1/3$ for the up- and down-type components.

Because $U(1)_Y$ is Abelian, the corresponding field-strength tensor is

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu. \quad (2.24)$$

There is no self-interaction term involving two B_μ fields in $B_{\mu\nu}$. Although $U(1)_Y$ is Abelian, it should not be identified with the electromagnetic gauge symmetry $U(1)_{\text{em}}$. Before electroweak symmetry breaking, the corresponding gauge field is the hypercharge field B_μ rather than the photon field. The electromagnetic gauge symmetry appears only after electroweak symmetry breaking, when B_μ mixes with the neutral weak gauge field W_μ^3 to form the photon and the Z boson.

2.1.5 Covariant derivative and gauge field strengths

Combining the three gauge symmetries, the covariant derivative acting on a generic SM field can be written as

$$D_\mu = \partial_\mu - ig_s G_\mu^a T_C^a - ig W_\mu^I T_L^I - ig' Y B_\mu. \quad (2.25)$$

Here g_s , g , and g' are the gauge couplings of $SU(3)_C$, $SU(2)_L$, and $U(1)_Y$, respectively. The matrices T_C^a and T_L^I are the generators acting on the particular field. For a colour triplet, $T_C^a = \lambda^a/2$, while for a colour singlet $T_C^a = 0$. For an $SU(2)_L$ doublet, $T_L^I = \tau^I/2$, while

for an $SU(2)_L$ singlet $T_L^I = 0$. The value of Y is the hypercharge of the field, as listed in Table 2.1.

The gauge kinetic terms are constructed from the field-strength tensors:

$$G_{\mu\nu}^a = \partial_\mu G_\nu^a - \partial_\nu G_\mu^a + g_s f^{abc} G_\mu^b G_\nu^c, \quad (2.26)$$

$$W_{\mu\nu}^I = \partial_\mu W_\nu^I - \partial_\nu W_\mu^I + g \epsilon^{IJK} W_\mu^J W_\nu^K, \quad (2.27)$$

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu. \quad (2.28)$$

The tensors $G_{\mu\nu}^a$ and $W_{\mu\nu}^I$ contain non-linear terms because $SU(3)_C$ and $SU(2)_L$ are non-Abelian groups. By contrast, $B_{\mu\nu}$ has the Abelian form because $U(1)_Y$ has only one generator.

2.1.6 The Standard Model Lagrangian

The SM Lagrangian is constructed from the field content described above and is invariant under the local gauge transformations of $SU(3)_C \times SU(2)_L \times U(1)_Y$. Using the covariant derivative and field-strength tensors introduced in the previous subsections, the minimal SM Lagrangian can be written as the sum of gauge-boson kinetic terms, fermion kinetic terms, the Higgs-sector terms, and Yukawa interactions. Ignoring gauge-fixing terms, ghost terms, and the QCD θ -term, it is written schematically as

$$\mathcal{L}_{\text{SM}} = \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{Higgs}} + \mathcal{L}_{\text{Yukawa}}. \quad (2.29)$$

The four terms in Eq. (2.29) are fixed by Lorentz invariance, local gauge invariance, and renormalisability, once the SM field content is specified [5, 6].

The gauge-boson kinetic terms are

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4} G_{\mu\nu}^a G^{a\mu\nu} - \frac{1}{4} W_{\mu\nu}^I W^{I\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu}. \quad (2.30)$$

Here $G_{\mu\nu}^a$, $W_{\mu\nu}^I$, and $B_{\mu\nu}$ are the field-strength tensors of $SU(3)_C$, $SU(2)_L$, and $U(1)_Y$, respectively. The indices $a = 1, \dots, 8$ and $I = 1, 2, 3$ label the generators of $SU(3)_C$ and $SU(2)_L$, and repeated Lorentz and gauge indices are summed over. The common factor $-1/4$ gives the conventional canonical normalisation of the gauge-boson kinetic terms. With this normalisation, the quadratic parts of Eq. (2.30) describe the free propagation of the gluon, weak, and hypercharge gauge fields, while the non-Abelian parts of $G_{\mu\nu}^a$ and $W_{\mu\nu}^I$ generate self-interactions among the corresponding gauge bosons.

The fermion kinetic terms are

$$\mathcal{L}_{\text{fermion}} = \sum_{i=1}^3 \left[\bar{Q}_L^i i\gamma^\mu D_\mu Q_L^i + \bar{u}_R^i i\gamma^\mu D_\mu u_R^i + \bar{d}_R^i i\gamma^\mu D_\mu d_R^i + \bar{L}_L^i i\gamma^\mu D_\mu L_L^i + \bar{e}_R^i i\gamma^\mu D_\mu e_R^i \right]. \quad (2.31)$$

The sum runs over the three fermion generations. The fields Q_L^i and L_L^i are the left-handed quark and lepton doublets, while u_R^i , d_R^i , and e_R^i are right-handed singlets under $SU(2)_L$. The Dirac adjoint is defined as $\bar{\psi} = \psi^\dagger \gamma^0$, and γ^μ are the Dirac gamma matrices. The covariant derivative D_μ contains the gauge fields and couplings appropriate to the gauge quantum numbers of each fermion field. Therefore, Eq. (2.31) describes both the free propagation of fermions and their gauge interactions.

The Higgs-sector terms are

$$\mathcal{L}_{\text{Higgs}} = (D_\mu \Phi)^\dagger (D^\mu \Phi) - V(\Phi), \quad V(\Phi) = -\mu^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2. \quad (2.32)$$

The first term is the kinetic term of the Higgs doublet and also contains the gauge interactions of the Higgs field through the covariant derivative. The function $V(\Phi)$ is the Higgs potential. With the convention used here, $\mu^2 > 0$ and $\lambda > 0$. The condition $\lambda > 0$ ensures that the potential is bounded from below, while the negative quadratic term causes the minimum of the potential to occur at a non-zero value of $\Phi^\dagger \Phi$. This leads to electroweak symmetry breaking through the Brout–Englert–Higgs mechanism [? 7?].

The Yukawa interactions are

$$\mathcal{L}_{\text{Yukawa}} = -y_d^{ij} \bar{Q}_L^i \Phi d_R^j - y_u^{ij} \bar{Q}_L^i \tilde{\Phi} u_R^j - y_e^{ij} \bar{L}_L^i \Phi e_R^j + \text{h.c.}, \quad (2.33)$$

where $i, j = 1, 2, 3$ are generation indices, and y_u^{ij} , y_d^{ij} , and y_e^{ij} are complex Yukawa coupling matrices in flavour space. h.c. denotes the Hermitian conjugate. The conjugate Higgs doublet is defined as

$$\tilde{\Phi} = i\tau^2 \Phi^*, \quad (2.34)$$

where τ^2 is the second Pauli matrix. The field $\tilde{\Phi}$ transforms as an $SU(2)_L$ doublet with hypercharge opposite to that of Φ , allowing a gauge-invariant Yukawa interaction for up-type quarks. After electroweak symmetry breaking, the Yukawa interactions generate masses for the charged fermions. In the minimal SM, no right-handed neutrino field is included, and therefore no renormalisable neutrino Yukawa mass term appears in Eq. (2.33).

2.1.7 Electroweak symmetry breaking and the Higgs mechanism

The Higgs potential in Eq. (2.32) has a minimum at a non-zero value of $\Phi^\dagger \Phi$. For the convention used in Eq. (2.32), the minimum satisfies $\Phi^\dagger \Phi = v^2/2$, where $v = \sqrt{\mu^2/\lambda}$ is the Higgs vacuum expectation value and sets the electroweak scale. A convenient choice of vacuum expectation value is

$$\langle \Phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}. \quad (2.35)$$

The value of v is related to the Fermi constant by $v = (\sqrt{2}G_F)^{-1/2} \simeq 246 \text{ GeV}$ [8].

This vacuum breaks the electroweak symmetry according to

$$SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}. \quad (2.36)$$

The unbroken symmetry is electromagnetism. This mass-generation mechanism for gauge bosons is the Brout–Englert–Higgs mechanism [? 7?].

In unitary gauge, the Higgs doublet can be written as

$$\Phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \quad (2.37)$$

where $h(x)$ is the physical Higgs boson field. Expanding the Higgs potential around the vacuum gives

$$V(\Phi) = V(\langle \Phi \rangle) + \frac{1}{2}m_h^2 h^2 + \lambda v h^3 + \frac{\lambda}{4}h^4, \quad (2.38)$$

where the physical Higgs-boson mass is

$$m_h^2 = 2\lambda v^2 = 2\mu^2. \quad (2.39)$$

Unlike the W and Z boson masses, which are fixed at tree level by the electroweak gauge couplings and the vacuum expectation value, the Higgs-boson mass depends on the quartic coupling λ . Therefore, the SM does not predict the numerical value of m_h before λ is fixed experimentally. The Higgs boson was observed by the ATLAS and CMS experiments in 2012 [9, 10]. The observed Higgs-boson mass is approximately 125 GeV, corresponding to $\lambda \simeq m_h^2/(2v^2) \simeq 0.13$ [11].

The masses of the electroweak gauge bosons arise from the Higgs kinetic term,

$$(D_\mu \Phi)^\dagger (D^\mu \Phi), \quad (2.40)$$

after substituting the Higgs vacuum expectation value. It is useful to define the charged weak fields as

$$W_\mu^\pm = \frac{1}{\sqrt{2}} (W_\mu^1 \mp iW_\mu^2). \quad (2.41)$$

The terms quadratic in the electroweak gauge fields are then

$$(D_\mu \langle \Phi \rangle)^\dagger (D^\mu \langle \Phi \rangle) = \frac{g^2 v^2}{4} W_\mu^+ W^{-\mu} + \frac{v^2}{8} (gW_\mu^3 - g'B_\mu)^2. \quad (2.42)$$

The first term is the mass term of the charged weak bosons. The second term shows that the neutral gauge fields W_μ^3 and B_μ mix and must be diagonalised.

The weak mixing angle is defined by

$$\cos \theta_W = \frac{g}{\sqrt{g^2 + g'^2}}, \quad \sin \theta_W = \frac{g'}{\sqrt{g^2 + g'^2}}. \quad (2.43)$$

The corresponding neutral mass eigenstates are

$$A_\mu = B_\mu \cos \theta_W + W_\mu^3 \sin \theta_W, \quad (2.44)$$

$$Z_\mu = -B_\mu \sin \theta_W + W_\mu^3 \cos \theta_W. \quad (2.45)$$

The field A_μ is identified with the photon, while Z_μ is the neutral weak boson. The photon remains massless because it is associated with the unbroken gauge symmetry $U(1)_{\text{em}}$.

The resulting gauge-boson masses are

$$m_W = \frac{1}{2} g v, \quad m_Z = \frac{1}{2} \sqrt{g^2 + g'^2} v = \frac{m_W}{\cos \theta_W}, \quad m_\gamma = 0. \quad (2.46)$$

Thus, the SM predicts the relation $m_W = m_Z \cos \theta_W$. Inserting $v \simeq 246$ GeV and the electroweak couplings gives $m_W \simeq 80$ GeV and $m_Z \simeq 91$ GeV. Therefore, the Higgs mechanism gives masses to the W and Z bosons while leaving the photon massless.

After electroweak symmetry breaking, the unbroken gauge symmetry is $U(1)_{\text{em}}$, whose gauge field is the photon field A_μ . At energies much below the electroweak scale, the electromagnetic interactions of charged fermions are described by quantum electrodynamics (QED).

For a charged fermion field ψ with electric charge Q , the QED Lagrangian can be written as

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu D_\mu - m)\psi, \quad (2.47)$$

where

$$D_\mu = \partial_\mu - ieQA_\mu, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (2.48)$$

Here e is the electromagnetic coupling constant, Q is the electric charge of the fermion in units of e , and $F_{\mu\nu}$ is the electromagnetic field-strength tensor. Thus, QED appears as the low-energy gauge theory associated with the unbroken $U(1)_{\text{em}}$ symmetry of the SM.

2.1.8 Fermion masses and flavour mixing

The Yukawa interactions generate fermion masses after electroweak symmetry breaking. When the Higgs field acquires its vacuum expectation value v , the Yukawa coupling matrices are converted into fermion mass matrices:

$$M_u^{ij} = \frac{v}{\sqrt{2}}y_u^{ij}, \quad M_d^{ij} = \frac{v}{\sqrt{2}}y_d^{ij}, \quad M_e^{ij} = \frac{v}{\sqrt{2}}y_e^{ij}. \quad (2.49)$$

Here M_u , M_d , and M_e are the mass matrices of the up-type quarks, down-type quarks, and charged leptons, respectively. The matrices y_u , y_d , and y_e are the corresponding dimensionless Yukawa coupling matrices, and the indices $i, j = 1, 2, 3$ label the fermion generations.

The physical fermion masses are obtained by diagonalising the mass matrices. In the quark sector, the up-type and down-type mass matrices are diagonalised by different rotations in flavour space. As a result, flavour mixing appears in the charged-current weak interaction. This mixing is described by the Cabibbo–Kobayashi–Maskawa (CKM) matrix [12, 13]. In the mass eigenstate basis, the charged-current interaction of quarks is

$$\mathcal{L}_{\text{CC}} = \frac{g}{\sqrt{2}}\bar{u}_L^i \gamma^\mu V_{ij}^{\text{CKM}} d_L^j W_\mu^+ + \text{h.c.} \quad (2.50)$$

Here g is the $SU(2)_L$ gauge coupling, u_L^i and d_L^j are the left-handed up-type and down-type quark mass eigenstates, and W_μ^+ is the charged weak gauge boson. The matrix element V_{ij}^{CKM} describes the strength of the charged-current transition from the down-type quark d^j to the up-type quark u^i . The matrices γ^μ are the Dirac gamma matrices, and the Hermitian-conjugate term describes the corresponding interaction involving W_μ^- . The CKM matrix is essential for describing flavour-changing charged-current processes such as $b \rightarrow c\tau\nu$, which

are relevant for the leptoquark scenarios discussed later. Its complex phase also provides a source of CP violation in the quark sector of the SM.

In the minimal SM, neutrino masses are absent because no right-handed neutrino fields are included and no renormalisable neutrino Yukawa interaction can be written with the minimal field content. This provides one of the motivations for considering extensions of the SM.

2.2 Beyond the Standard Model

Although the SM has been tested with high precision and has successfully described a wide range of experimental observations, it is not a complete theory of nature. Several experimental and theoretical facts point to physics beyond the SM.

2.2.1 Gravity and the Planck scale

The SM describes the electromagnetic, weak, and strong interactions as quantum gauge field theories, but it does not include gravity. Gravity is instead described at macroscopic scales by general relativity. A complete theory of nature is therefore expected to provide a consistent quantum description of gravity and to explain its connection to the SM interactions.

The difficulty of connecting gravity with the SM appears in two related ways. First, quantum gravitational effects are expected to become important at the Planck scale, $M_{\text{Pl}} \sim 10^{19}$ GeV. This is much higher than the electroweak scale, $v \simeq 246$ GeV. This large separation of scales is closely related to the hierarchy problem discussed below.

Second, constructing a quantum theory of gravity is theoretically challenging. If general relativity is quantised perturbatively in the same way as ordinary quantum field theories, it is not renormalisable as a fundamental theory. This makes it difficult to combine gravity with the quantum-field-theoretic framework of the SM in a straightforward way. Several possible approaches to quantum gravity have been proposed, including string theory, loop quantum gravity, and asymptotic safety, although no complete theory has been experimentally confirmed.

2.2.2 The hierarchy problem

The electroweak scale is set by the Higgs vacuum expectation value, $v \simeq 246$ GeV, and is closely related to the masses of the weak gauge bosons and the Higgs boson. In the SM, the

Higgs-boson mass is determined at tree level by

$$m_h^2 = 2\lambda v^2, \quad (2.51)$$

where λ is the Higgs quartic coupling. The observed value $m_h \simeq 125$ GeV therefore corresponds to $\lambda \simeq 0.13$.

The hierarchy problem arises because the Higgs mass parameter is highly sensitive to physics at very high energy scales. If the SM is treated as an effective theory valid up to an ultraviolet cutoff scale Λ , loop corrections to the Higgs mass can be written schematically as

$$m_{h,\text{phys}}^2 = m_{h,0}^2 + \delta m_h^2, \quad (2.52)$$

where $m_{h,0}^2$ is the bare Higgs mass parameter and δm_h^2 denotes the quantum correction. At one-loop order, the correction contains terms quadratically sensitive to the cutoff scale,

$$\delta m_h^2 \sim \frac{\Lambda^2}{16\pi^2}, \quad (2.53)$$

If the cutoff scale is close to the Planck scale, $\Lambda \sim M_{\text{Pl}} \sim 10^{19}$ GeV, the magnitude of δm_h^2 is many orders of magnitude larger than the observed value $m_h^2 \sim (125 \text{ GeV})^2$. Therefore, the bare parameter $m_{h,0}^2$ and the quantum correction δm_h^2 must cancel each other with extremely high precision in order to leave the observed electroweak-scale Higgs mass. This required cancellation is known as fine-tuning.

The hierarchy problem is the question of why the electroweak scale remains so much smaller than a possible ultraviolet scale such as the Planck scale. It motivates many extensions of the SM, such as supersymmetry, composite Higgs models, and other scenarios in which new physics appears near the TeV scale and stabilises the Higgs mass against large ultraviolet corrections. More generally, the hierarchy problem provides one of the motivations for experimental searches for new particles at the TeV scale.

2.2.3 Neutrino masses

In the minimal SM, neutrinos are massless because no right-handed neutrino fields are included and no renormalisable neutrino mass term can be written using only the SM field content. However, neutrino oscillation experiments have shown that neutrinos change flavour as they propagate, which requires non-zero neutrino mass differences. Evidence for atmospheric neutrino oscillations was reported by the Super-Kamiokande experiment [14], and solar neutrino flavour transformation was established by the Sudbury Neutrino Obser-

vatory [15]. Therefore, neutrino masses provide direct experimental evidence for physics beyond the minimal SM.

2.2.4 Dark matter

Astrophysical and cosmological observations indicate the existence of non-luminous matter that cannot be explained by the known particles of the SM. Measurements of the cosmic microwave background, for example by the Planck satellite, are well described by a cosmological model containing a significant non-baryonic dark matter component [16]. The SM does not contain a suitable cold dark matter candidate. Although neutrinos are electrically neutral and weakly interacting, their small masses and relativistic behaviour in the early universe do not provide the required cold dark matter component.

2.2.5 Matter–antimatter asymmetry

The observable universe contains much more matter than antimatter. Sakharov identified three necessary conditions for dynamically generating such a baryon asymmetry: baryon-number violation, C and CP violation, and departure from thermal equilibrium [17]. The SM contains CP violation through the CKM matrix and baryon-plus-lepton number violation through non-perturbative electroweak effects. However, the amount of CP violation in the SM and the nature of the electroweak phase transition are not sufficient to explain the observed baryon asymmetry of the universe [18, 19]. This motivates new sources of CP violation and new dynamics in the early universe.

2.2.6 The strong CP problem

The most general QCD Lagrangian allows a CP-violating term of the form

$$\mathcal{L}_\theta = \theta \frac{g_s^2}{32\pi^2} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}, \quad \tilde{G}^{a\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\rho\sigma} G_{\rho\sigma}^a. \quad (2.54)$$

Here g_s is the strong coupling constant, and $G_{\mu\nu}^a$ is the gluon field-strength tensor of the $SU(3)_C$ gauge group, with the colour index $a = 1, \dots, 8$ labelling the eight gluon fields. The tensor $\tilde{G}^{a\mu\nu}$ is the dual field-strength tensor, defined using the totally antisymmetric Levi-Civita tensor $\varepsilon^{\mu\nu\rho\sigma}$. The parameter θ is a dimensionless QCD vacuum angle that controls the size of this CP-violating interaction. The factor $g_s^2/(32\pi^2)$ is a conventional normalisation.

The operator $G_{\mu\nu}^a \tilde{G}^{a\mu\nu}$ is gauge invariant and has mass dimension four, so it is allowed by the gauge symmetry and renormalisability of QCD. However, it violates CP symmetry. Therefore, a non-zero value of the effective QCD vacuum angle would lead to CP violation in the strong interaction.

This term would contribute to the electric dipole moment of the neutron. Experimental limits on the neutron electric dipole moment strongly constrain the effective QCD vacuum angle $\bar{\theta}$ [20]. The effective parameter $\bar{\theta}$ includes both the QCD vacuum angle θ and phases from the quark mass matrix. The absence of observable strong CP violation, despite the fact that such a term is allowed by the SM gauge symmetries, is known as the strong CP problem.

2.2.7 Flavour anomalies and lepton flavour universality

The flavour structure of the SM is mainly determined by the Yukawa matrices. After electroweak symmetry breaking, these matrices generate the fermion mass matrices; their eigenvalues determine the fermion masses, while the mismatch between the diagonalisation of the up- and down-type quark mass matrices gives rise to quark flavour mixing through the CKM matrix.

Although the SM accommodates the observed masses and mixing angles, it does not explain their hierarchical pattern or their origin. Precision measurements in flavour physics therefore provide sensitive probes of possible physics beyond the SM.

A key property of the SM is lepton flavour universality (LFU), which means that the electroweak gauge interactions couple equally to the three charged-lepton generations, apart from effects due to their different masses. Semileptonic B -meson decays provide precise tests of LFU by comparing decay rates involving different charged leptons. In charged-current transitions of the form $b \rightarrow c\ell\nu$, the commonly studied ratios are

$$R(D^{(*)}) = \frac{\mathcal{B}(\bar{B} \rightarrow D^{(*)} \tau \bar{\nu}_\tau)}{\mathcal{B}(\bar{B} \rightarrow D^{(*)} \ell \bar{\nu}_\ell)}, \quad \ell = e, \mu. \quad (2.55)$$

These observables compare decays involving a τ lepton with the corresponding decays involving light charged leptons.

Neutral-current transitions of the form $b \rightarrow s\ell^+\ell^-$ can be tested through ratios such as

$$R(K^{(*)}) = \frac{\mathcal{B}(B \rightarrow K^{(*)} \mu^+ \mu^-)}{\mathcal{B}(B \rightarrow K^{(*)} e^+ e^-)}. \quad (2.56)$$

These ratios compare final states with muon pairs and electron pairs. Recent measurements of $R(K)$ and $R(K^*)$ are compatible with SM predictions [21, 22]. However, measurements of

$R(D)$ and $R(D^*)$ continue to show tensions with SM predictions. The HFLAV CKM 2025 average gives a combined discrepancy of about 3.8σ with respect to the SM prediction [23] as shown in Figure 2.1. These observables remain important motivations for models that couple differently to different lepton generations.

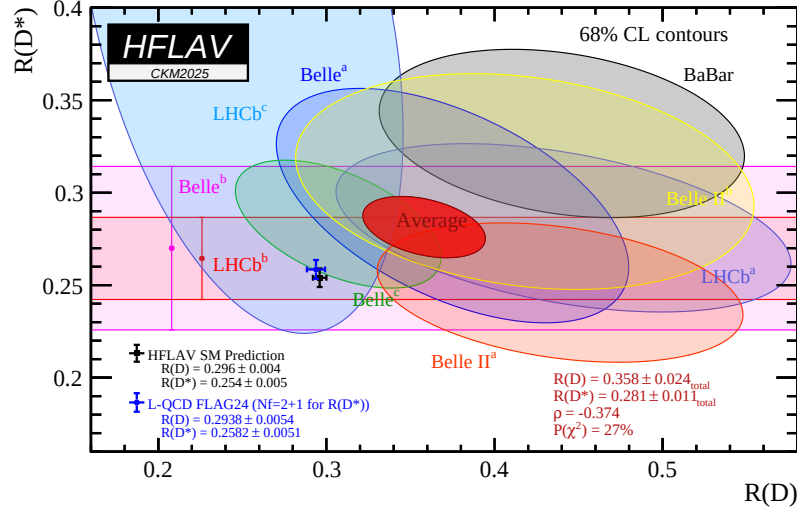


Fig. 2.1 Average of $R(D^*)$ and $R(D)$ [23] measured by the BaBar [24, 25], Belle [26–29], Belle-II [30, 31], and LHCb [32–34] experiments.

2.3 Possible Solutions to the Remaining Problems of the Standard Model

Many extensions of the SM have been proposed to address one or more of the problems described above. This section briefly introduces several representative ideas: supersymmetry, axions, and leptiquarks. These models are not mutually exclusive, and in many cases they address different aspects of the SM's limitations.

2.3.1 Supersymmetry

Supersymmetry (SUSY) is a spacetime symmetry relating bosons and fermions [35, 36]. In supersymmetric theories, each SM particle is accompanied by a superpartner whose spin differs by one half unit. The fermionic superpartners of bosons are conventionally named by adding the suffix ‘-ino’, such as the gluino, wino, bino, and higgsino, while the bosonic superpartners of fermions are named by adding the prefix ‘s-’, such as squarks and sleptons.

If supersymmetry were exact, particles and their superpartners would have equal masses. Since no such superpartners have been observed, SUSY must be broken if it is realised in nature.

One of the main motivations for SUSY is the hierarchy problem. Loop corrections to the Higgs mass from SM particles can be cancelled by corresponding contributions from their superpartners, reducing the sensitivity of the electroweak scale to very high energy scales [37, 36]. In models with conserved R -parity, supersymmetric particles are produced in pairs and the lightest supersymmetric particle (LSP) is stable [38, 36]. If the LSP is electrically neutral and weakly interacting, such as a neutralino in many SUSY scenarios, it can provide a dark matter candidate [39, 40]. SUSY can also improve the unification of gauge couplings at high energy, since the additional superpartners modify the renormalisation-group evolution of the gauge couplings [41, 42]. However, no direct evidence for supersymmetric particles has yet been observed, and the simplest low-scale SUSY scenarios are increasingly constrained by collider searches [43].

2.3.2 Axions and the strong CP problem

The Peccei–Quinn mechanism provides a dynamical solution to the strong CP problem [44, 45]. It introduces a new global chiral symmetry, usually denoted as the Peccei–Quinn symmetry $U(1)_{\text{PQ}}$, which is spontaneously broken. The associated pseudo-Nambu–Goldstone boson is the axion [46, 47]. The axion field dynamically relaxes the effective value of $\bar{\theta}$ to zero, thereby explaining the absence of observable strong CP violation. In many models, axions or axion-like particles can also contribute to the dark matter density of the universe [48].

2.3.3 Leptoquarks

Leptoquarks provide another well-motivated class of particles beyond the Standard Model. They are hypothetical bosons that couple directly to a quark and a lepton, thereby connecting the quark and lepton sectors. Such particles appear naturally in several extensions of the SM, including models based on quark–lepton unification, such as the Pati–Salam framework [49].

Leptoquarks are particularly interesting because they can affect semileptonic processes. This makes them relevant to the flavour anomalies observed in several B -meson decay measurements, where the experimental results have shown tensions with the SM predictions in observables testing lepton-flavour universality.

Among the possible leptoquark scenarios, the vector singlet leptoquark U_1 is especially well motivated for charged-current anomalies such as $b \rightarrow c \tau \nu$. This motivates collider

searches for final states containing hadronically decaying τ leptons. The following chapter introduces the leptoquark models in more detail and the leptoquark searches at the LHC.

Chapter 3

Leptoquark Search at the LHC

3.1 Leptoquark Models and Phenomenology

3.1.1 General properties of leptoquarks

Leptoquarks (LQs) are hypothetical colour-charged bosons that couple directly to a quark and a lepton and therefore mediate interactions connecting the quark and lepton sectors. Depending on their spin, leptoquarks can be classified as scalar leptoquarks with spin zero or vector leptoquarks with spin one. They appear in several theoretical frameworks, including grand unified theories such as $SU(5)$ and $SO(10)$, quark–lepton unification models such as the Pati–Salam model, composite models, and models motivated by flavour anomalies [50, 49, 51].

Leptoquarks are particularly interesting in the context of semileptonic flavour physics. Their tree-level exchange can generate effective four-fermion interactions involving two quark fields and two lepton fields. Such interactions can modify flavour observables in processes such as $b \rightarrow c \tau \nu$ and $b \rightarrow s \ell^+ \ell^-$. In addition, the flavour structure of LQ couplings is model dependent and can be arranged to favour interactions with third-generation fermions. This feature is motivated by the distinctive role of the top quark, bottom quark, and τ -lepton in many flavour-motivated scenarios.

At the LHC, leptoquarks can be searched for directly. Since they carry colour charge, they can be pair-produced through QCD interactions. Additional production modes, such as single production or non-resonant exchange, are also possible, depending on the LQ coupling to SM quarks and leptons. Therefore, the LQ searches in ATLAS provide information complementary to low-energy flavour measurements: flavour observables probe virtual effects through precision measurements, while the search at pp collider probes direct LQ production or high-energy virtual exchange.

3.1.2 The simplified U_1 leptoquark model

Several leptoquark representations have been studied in connection with semileptonic B -meson anomalies, including the scalar singlet S_1 and the scalar doublet R_2 . In this thesis, however, the benchmark interpretation is based on the vector singlet leptoquark U_1 . The U_1 benchmark is well motivated because its left-handed coupling to quark and lepton doublets can generate the SM-like vector operator relevant to $b \rightarrow c \tau \nu$. In addition, flavour structures with enhanced couplings to third-generation fermions naturally lead to final states involving τ -leptons, neutrinos, and jets.

The U_1 leptoquark transforms as $U_1^\mu \sim (3, 1, 2/3)$ under the SM gauge group. It is therefore a colour-triplet, weak-singlet vector boson with hypercharge $Y = 2/3$. Since it is an $SU(2)$ singlet, its electric charge is $Q = T^3 + Y = 2/3$.

A commonly used simplified-model Lagrangian [52] for U_1 is

$$\begin{aligned} \mathcal{L}_U = & -\frac{1}{2}U_{1\mu\nu}^\dagger U_1^{\mu\nu} + M_U^2 U_{1\mu}^\dagger U_1^\mu \\ & -ig_c(1 - \kappa_c)U_{1\mu}^\dagger T^a U_{1\nu} G^{a\mu\nu} - i\frac{2}{3}g_Y(1 - \kappa_Y)U_{1\mu}^\dagger U_{1\nu} B^{\mu\nu} \\ & + (U_1^\mu J_\mu + \text{h.c.}). \end{aligned} \quad (3.1)$$

Here U_1^μ is the vector leptoquark field and M_U is the leptoquark mass. The first term is the kinetic term, while the second term is the mass term. The field-strength tensor of U_1 is defined as

$$U_{1\mu\nu} = D_\mu U_{1\nu} - D_\nu U_{1\mu}, \quad (3.2)$$

with the covariant derivative

$$D_\mu = \partial_\mu - ig_c G_\mu^a T^a - i\frac{2}{3}g_Y B_\mu. \quad (3.3)$$

Here g_c and g_Y are the colour and hypercharge gauge couplings. The fields G_μ^a and B_μ are the gluon and hypercharge gauge fields, and $G^{a\mu\nu}$ and $B^{\mu\nu}$ are their corresponding field-strength tensors. The matrices T^a are the generators of $SU(3)$. For a colour-triplet field, they are given by $T^a = \lambda^a/2$, where λ^a are the Gell-Mann matrices and $a = 1, \dots, 8$.

The parameters κ_c and κ_Y control the interactions of the vector leptoquark with the SM gauge fields. $\kappa_c = \kappa_Y = 0$ corresponds to the Yang–Mills scenario, whereas $\kappa_c = \kappa_Y = 1$ corresponds to the minimal-coupling scenario [52, 53]. The Yang–Mills scenario is adopted throughout this thesis.

The last term in Eq. (3.1) describes the interaction of U_1 with SM fermions through the current J_μ . It is given by

$$J_\mu = \frac{g_U}{\sqrt{2}} [\beta_L^{i\alpha} (\bar{q}_L^i \gamma_\mu \ell_L^\alpha) + \beta_R^{i\alpha} (\bar{d}_R^i \gamma_\mu e_R^\alpha)]. \quad (3.4)$$

Here $i = 1, 2, 3$ labels the quark generation and $\alpha = 1, 2, 3$ labels the lepton generation. q_L^i and ℓ_L^α are left-handed quark and lepton doublets, while d_R^i and e_R^α are right-handed down-type quark and charged-lepton singlets. The parameter g_U denotes the overall strength of the U_1 coupling to fermions. The matrices $\beta_L^{i\alpha}$ and $\beta_R^{i\alpha}$ are complex 3×3 flavour-coupling matrices for the left- and right-handed interactions, respectively. The strength of a given flavour coupling is therefore proportional to $g_U \beta_{L,R}^{i\alpha}$.

The down-quark and charged-lepton mass eigenstate basis is adopted for the $SU(2)$ multiplets [52]:

$$q_L^i = \begin{pmatrix} V_{ji}^* u_L^j \\ d_L^i \end{pmatrix}, \quad \ell_L^\alpha = \begin{pmatrix} \nu_L^\alpha \\ e_L^\alpha \end{pmatrix}, \quad (3.5)$$

where V is the CKM matrix.

Expanding the left-handed current from Eq. 3.4 gives

$$\bar{q}_L^i \gamma_\mu \ell_L^\alpha = V_{ji} \bar{u}_L^j \gamma_\mu \nu_L^\alpha + \bar{d}_L^i \gamma_\mu e_L^\alpha. \quad (3.6)$$

This expression shows that a single left-handed flavour coupling $\beta_L^{i\alpha}$ gives two related interactions. One involves an up-type quark together with a neutrino, while the other involves a down-type quark together with a charged lepton.

The simplified U_1 model allows general flavour couplings to different quark and lepton generations [52]. Meanwhile, we focus on the charged-current $b \rightarrow c \tau \nu$ transition and the signatures containing a τ -lepton and a neutrino. Therefore, the couplings to second-generation leptons in our benchmark are set to zero, which is $\beta_L^{22} = \beta_L^{32} = 0$, and only the couplings involving third-generation leptons are retained. Finally, the flavour-coupling matrices are taken to be

$$\beta_L = \begin{pmatrix} 0 & 0 & \beta_L^{13} \\ 0 & 0 & \beta_L^{23} \\ 0 & 0 & \beta_L^{33} \end{pmatrix}, \quad \beta_R = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \beta_R^{33} \end{pmatrix}. \quad (3.7)$$

The third-generation coupling is normalised to $\beta_L^{33} = 1$, while g_U remains a free parameter controlling the overall fermionic coupling strength.

Within this benchmark, β_L^{23} controls the $s\tau$ and $c\nu_\tau$ coupling through the CKM factor V_{cs} , while β_L^{33} controls the $b\tau$ coupling and contributes $c\nu_\tau$ coupling as $V_{cb}\beta_L^{33}$.

The heavy U_1 field can be integrated out at the matching scale $\mu \sim M_U$. At tree level, its exchange generates four-fermion operators involving left- and right-handed quark and lepton fields [54]. The corresponding effective Lagrangian can be written as

$$\mathcal{L}_{\text{eff}}^U = -\frac{2}{v^2} \left[C_{LL}^{ij\alpha\beta} \mathcal{O}_{LL}^{ij\alpha\beta} + C_{RR}^{ij\alpha\beta} \mathcal{O}_{RR}^{ij\alpha\beta} + \left(C_{LR}^{ij\alpha\beta} \mathcal{O}_{LR}^{ij\alpha\beta} + \text{h.c.} \right) \right]. \quad (3.8)$$

The four-fermion operators are defined as

$$\mathcal{O}_{LL}^{ij\alpha\beta} = (\bar{q}_L^i \gamma_\mu \ell_L^\alpha) (\bar{\ell}_L^\beta \gamma^\mu q_L^j), \quad (3.9)$$

$$\mathcal{O}_{RR}^{ij\alpha\beta} = (\bar{d}_R^i \gamma_\mu e_R^\alpha) (\bar{e}_R^\beta \gamma^\mu d_R^j), \quad (3.10)$$

$$\mathcal{O}_{LR}^{ij\alpha\beta} = (\bar{q}_L^i \gamma_\mu \ell_L^\alpha) (\bar{e}_R^\beta \gamma^\mu d_R^j). \quad (3.11)$$

The Wilson coefficients are

$$C_{LL}^{ij\alpha\beta} = C_U \beta_L^{i\alpha} (\beta_L^{j\beta})^*, \quad (3.12)$$

$$C_{RR}^{ij\alpha\beta} = C_U \beta_R^{i\alpha} (\beta_R^{j\beta})^*, \quad (3.13)$$

$$C_{LR}^{ij\alpha\beta} = C_U \beta_L^{i\alpha} (\beta_R^{j\beta})^*. \quad (3.14)$$

The overall coefficient is defined as

$$C_U = \frac{g_U^2 v^2}{4M_U^2}, \quad (3.15)$$

where $v \simeq 246$ GeV is the Higgs vacuum expectation value. The coefficient C_U sets the overall strength of the effective interaction and scales as g_U^2/M_U^2 , while the flavour dependence is encoded in the coupling matrices β_L and β_R .

The effective Lagrangian for the $b \rightarrow c\tau\nu$ transition can be written as:

$$\begin{aligned} \mathcal{L}_{\text{eff}}^{b \rightarrow c\tau\nu} = & -\frac{4G_F}{\sqrt{2}} V_{cb} \left[(1 + C_{LL}^c) (\bar{c}_L \gamma_\mu b_L) (\bar{\tau}_L \gamma^\mu \nu_{\tau L}) \right. \\ & \left. - 2C_{LR}^c (\bar{c}_L b_R) (\bar{\tau}_R \nu_{\tau L}) \right], \end{aligned} \quad (3.16)$$

where G_F is the Fermi constant. The coefficient C_{LL}^c modifies the SM-like left-handed vector interaction, while C_{LR}^c is associated with the scalar operator arising from the combination of left- and right-handed fermion couplings of the U_1 leptoquark.

For the flavour structure introduced above, the corresponding coefficients are

$$C_{LL}^c = C_U \frac{(\beta_L^{33})^*}{V_{cb}} (V_{cd}\beta_L^{13} + V_{cs}\beta_L^{23} + V_{cb}\beta_L^{33}), \quad (3.17)$$

$$C_{LR}^c = C_U \frac{(\beta_R^{33})^*}{V_{cb}} (V_{cd}\beta_L^{13} + V_{cs}\beta_L^{23} + V_{cb}\beta_L^{33}), \quad (3.18)$$

Using the normalisation $\beta_L^{33} = 1$ and neglecting the small first-generation coupling β_L^{13} , the coefficients reduce to¹

$$C_{LL}^c \simeq C_U \left(1 + \frac{V_{cs}}{V_{cb}} \beta_L^{23} \right), \quad (3.19)$$

$$C_{LR}^c \simeq C_U (\beta_R^{33})^* \left(1 + \frac{V_{cs}}{V_{cb}} \beta_L^{23} \right). \quad (3.20)$$

3.1.3 Low-energy fit and benchmark scenarios

The Wilson coefficients C_{LL}^c and C_{LR}^c introduced above can be constrained using measurements of semileptonic charged-current B -meson decays. Ref. [54] performed a combined fit to low-energy $b \rightarrow c \tau \nu$ observables and determines the preferred region in the (C_{LL}^c, C_{LR}^c) plane. The fit result is shown in Fig. 3.1. The Wilson coefficients in this figure are evolved to a common reference scale of 1 TeV to facilitate comparison with high-energy observables. The best-fit point is $(C_{LL}^c, C_{LR}^c) = (0.05, -0.02)$, and the preferred region is compatible with both pure left-handed ($\beta_R^{33} = 0$) and sizeable right-handed ($\beta_R^{33} = -1$) scenarios [54].

Pure left-handed scenario. For $\beta_R^{33} = 0$, Eq. 3.20 gives $C_{LR}^c = 0$. Therefore, the U_1 leptoquark contribution to $b \rightarrow c \tau \nu$ is described only by the left-handed vector operator, as given by Eq. 3.19. This is the primary benchmark scenario considered in this thesis. The corresponding B -anomaly-favoured region can be obtained from the intersection of the low-energy fit with the line $C_{LR}^c = 0$ in Fig. 3.1.

Right-handed scenario. The second benchmark is obtained by allowing a sizeable right-handed third-generation coupling. For $\beta_R^{33} = -1$, the relation between the Wilson coefficients

¹The small first-generation coupling $\beta_L^{13} = (V_{td}^*/V_{ts}^*)\beta_L^{23} \sim 0.2\beta_L^{23}$.

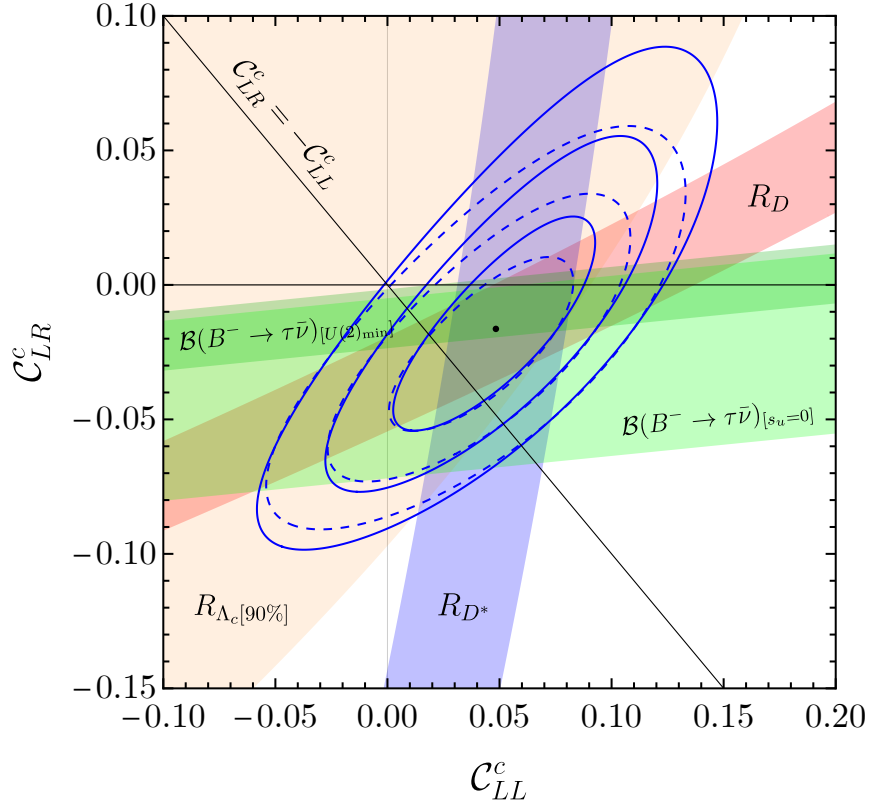


Fig. 3.1 The result in (C_{LL}^c, C_{LR}^c) plane obtained from the fit to low-energy $b \rightarrow c \tau \nu$ observables. The blue solid ellipses correspond to the 1, 2 and 3 σ contours. The blue dotted lines are obtained including also the constraint from $\mathcal{B}(B_u^- \rightarrow \tau \bar{\nu})$. The black point indicates the best-fit point of the fit, $(C_{LL}^c, C_{LR}^c) = (0.05, -0.02)$. The coloured bands show the $\Delta\chi^2 = 1$ regions preferred by the individual observables. The pure left-handed scenario ($\beta_R^{33} = 0$) and the equal magnitude right-handed scenario ($\beta_R^{33} = -1$) are given by the black solid lines [54].

becomes

$$C_{LR}^c = -C_{LL}^c. \quad (3.21)$$

This benchmark therefore corresponds to the diagonal line in Fig. 3.1, where the right-handed third-generation coupling has the same magnitude as the left-handed coupling.

The two scenarios are described by the parameter space (β_L^{23}, M_U, g_U) with different fixed values of β_R^{33} . Particular emphasis is placed on β_L^{23} , since charged-current low-energy data alone do not provide a stringent constraint on this coupling. For fixed M_U and g_U , β_L^{23} has a strong impact on C_{LL}^c because its contribution is enhanced by the CKM ratio V_{cs}/V_{cb} .

Indeed, one can constrain the size of β_L^{23} by considering possible ultraviolet (UV) completions of the simplified U_1 model. While the simplified model parametrises the relevant interactions directly in terms of the couplings β_L^{ij} , the UV completion provides a more fundamental description of their origin. One example is the “4321” model, which is based on the extended gauge group $\mathcal{G}_{4321} = SU(4) \times SU(3)' \times SU(2)_L \times U(1)'$. In the flavour structure considered in this model, the U_1 couplings are non-universal, with $|\beta_L^{23}| < |\beta_L^{33}|$ [52].

Additional constraints can be introduced when considering $\Delta F = 2$ observables. These are processes in which a quark-flavour quantum number changes by two units, including neutral-meson mixing such as $B_s - \bar{B}_s$, $K - \bar{K}$, and $D - \bar{D}$ mixing. In a given UV completion, the flavour-symmetry breaking that generates non-universal U_1 couplings may also be transmitted to the neutral-current sector, inducing flavour-changing neutral currents (FCNC) and consequently additional contributions to $\Delta F = 2$ processes. The strong experimental constraints on neutral-meson mixing therefore restrict the allowed flavour-violating structure of the UV model and indirectly constrain β_L^{23} .

Combining such considerations of the UV flavour structure with constraints from $\Delta F = 2$ observables, Ref. [54] identifies

$$\beta_L^{23} \in [0.06, 0.16] \quad (3.22)$$

as a theoretically motivated range for experimental searches. **However, this constraint should not be interpreted as a model-independent experimental bound on β_L^{23} .** Additional particles, flavour symmetries, or cancellation mechanisms may suppress flavour-changing transitions and thereby allow larger values of β_L^{23} . Therefore, large values of β_L^{23} remain a possibility within the U_1 model, and it is important to test this complementary region directly.

3.1.4 Leptoquark searches at the LHC

Leptoquarks can be searched for at the LHC through several production modes, including pair production, resonant single production, and non-resonant t -channel exchange. Pair production is dominated by QCD interactions. For a fixed κ choice, its cross section depends strongly on M_U but weakly on the couplings g_U and $\beta^{i\alpha}$. Pair-production searches therefore provide relatively robust constraints on the leptoquark mass.

ATLAS and CMS have performed searches for pair-produced leptoquarks coupling to third-generation fermions in final states containing τ -leptons, b -jets, and, depending on the decay mode, missing transverse momentum [55, 56]. For the vector U_1 , the mass limit extends to approximately 1.5–1.8 TeV as shown in Fig. 3.3. The limit also depends on the choice of the U_1 -gluon coupling, such as the Yang–Mills or minimal-coupling scenario, as well as on the decay pattern determined by the assumed fermionic couplings.

For single production, the cross section also depends strongly on the leptoquark–fermion couplings. Searches for singly produced leptoquarks decaying into $b\tau$ have been performed in the $b\tau\tau$ final state, where the leptoquark is produced in association with a τ -lepton as shown in Fig. 3.2a [55, 57]. In addition, virtual t -channel U_1 exchange can contribute to final states such as $\tau\tau$ or $\tau\nu$, enhancing the event yields in the high-mass region. ATLAS and CMS have therefore searched for non-resonant leptoquark contributions in high-mass $\tau\tau$ production [57–59]. These searches can remain sensitive to masses beyond the reach of the direct pair-production searches, particularly for large g_U , since the sensitivity is approximately governed by g_U/M_U .

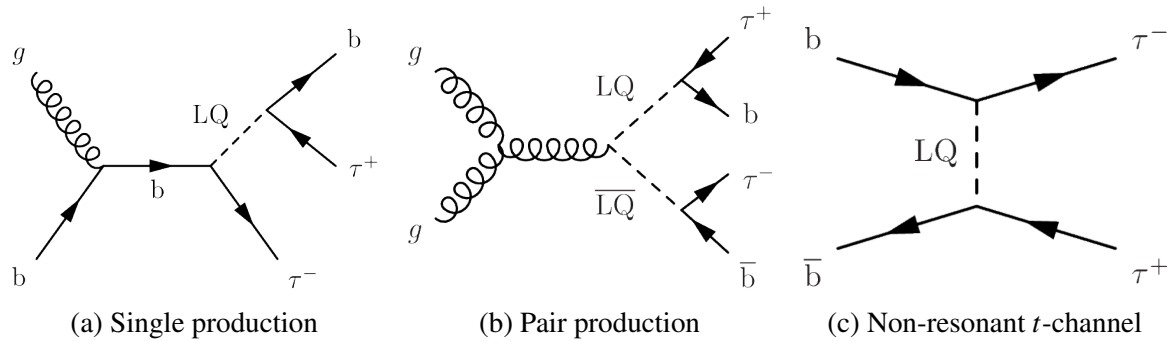


Fig. 3.2 Example Feynman diagrams of LQ production and exchange process for $\tau\tau(+b)$ searches.

On the other hand, the existing $\tau\tau(+b)$ searches generally assume β_L^{23} to be small or neglect its contribution, motivated by the flavour assumptions discussed in Sec. 3.1.3. Consequently, scenarios with a sizeable $c\nu_\tau$ coupling, and in particular the large- β_L^{23} region, have not been systematically explored at the LHC.

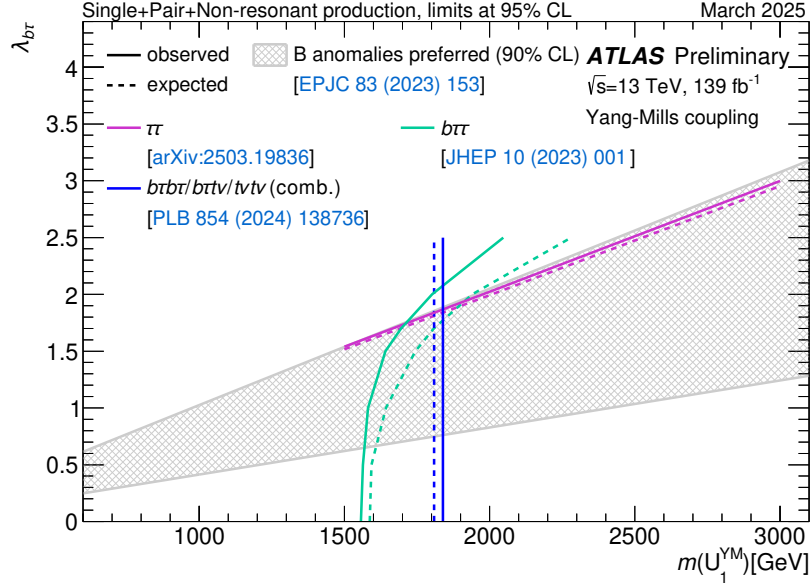


Fig. 3.3 Summary of the observed and expected 95% CL constraints on the U_1 vector-leptoquark coupling $\lambda_{b\tau}$ as a function of the leptoquark mass in the Yang–Mills coupling scenario, with $\beta_R^{33} = 0$. Pair-production searches dominate the direct mass constraint at lower masses, while single-production and $\tau\tau$ searches provide coupling-dependent sensitivity at larger masses. Here $\lambda_{b\tau} = g_U/\sqrt{2}$ for $\beta_L^{33} = 1$ [60].

The $\tau\nu$ final state provides a complementary probe of the $b\tau$ and $c\nu_\tau$ coupling. CMS has performed an inclusive search for events with a hadronically decaying τ -lepton and large missing transverse momentum, including an interpretation in terms of leptoquark exchange [61], but without an explicit b -jet requirement. This thesis presents a dedicated search in the $\tau\nu(+b)$ final state. Prior to the analysis, no dedicated ATLAS or CMS leptoquark search had targeted this topology. The $\tau\nu(+b)$ search presented here is therefore complementary to the existing $\tau\tau(+b)$ searches: it probes both the $b\tau$ and $c\nu_\tau$ couplings and extends the collider interpretation to large values of β_L^{23} . The corresponding benchmark and target signatures are introduced in the following subsection.

3.1.5 Signal model and target signatures

In this thesis, we focus on the simplified U_1 model introduced in Sec. 3.1.2 in the Yang–Mills scenario, corresponding to $\kappa_c = \kappa_Y = 0$ in Eq. 3.1. The third-generation coupling is normalised to $\beta_L^{33} = 1$, while the primary interpretation assumes a purely left-handed scenario ($\beta_R^{33} = 0$). An additional interpretation including a right-handed coupling is presented with the final results. Therefore, the signal is characterised by (M_U, g_U, β_L^{23}) .

Figure 3.4 shows representative U_1 signal diagrams targeted by the analysis. Both resonant single production and non-resonant t -channel exchange are considered, and each is divided according to the presence or absence of a b -tagged jet. This results in four target signatures: resonant $\tau\nu + b$, non-resonant $\tau\nu + b$, resonant $\tau\nu + 0b$, and non-resonant $\tau\nu + 0b$.

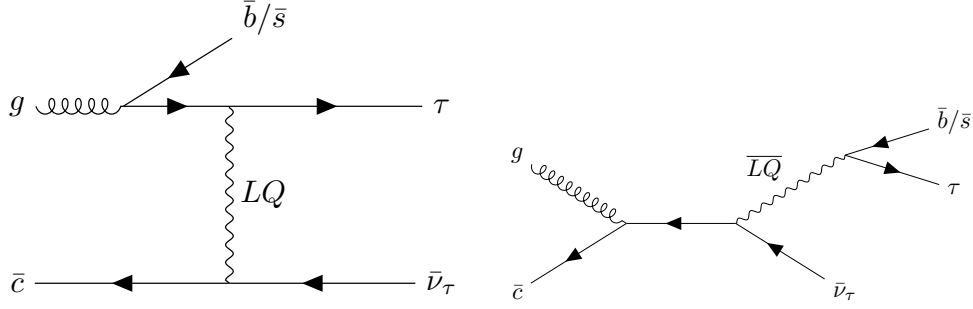


Fig. 3.4 Signal diagrams targeted in this analysis: non-resonant t -channel exchange (left) and resonant single production (right). The final-state jet can originate from a b - or s -quark.

The b -jet requirement provides strong discrimination against the dominant SM $\tau\nu + \text{jets}$ background, since genuine SM production of a $\tau\nu$ system in association with a b -quark is CKM suppressed [62]. The $\tau\nu + b$ categories are therefore particularly sensitive to the combination of the $b\tau$ and $c\nu_\tau$ interactions relevant to $b \rightarrow c\tau\nu$.

The zero- b categories extend the sensitivity to the large β_L^{23} region explored in this thesis. As β_L^{23} increases, the $s\tau$ interaction also becomes sizeable, producing signal events without a final-state b -quark. This contribution is particularly relevant for resonant production, where an on-shell U_1 can decay into $s\tau$. The four categories therefore provide complementary sensitivity to the different flavour and production configurations of the U_1 signal.

Chapter 4

LHC-ATLAS Experiment

4.1 Large Hadron Collider

The European Organisation for Nuclear Research (CERN) is an international particle-physics laboratory located near Geneva on the border between France and Switzerland. Since its foundation in 1954, CERN has developed and operated a chain of accelerators and experimental facilities to study the fundamental constituents of matter and their interactions [63]. The Large Hadron Collider (LHC) is the largest and highest-energy particle collider in the CERN accelerator complex. It is installed in the former Large Electron–Positron Collider (LEP) tunnel, which has a circumference of about 27 km and lies between approximately 45 m and 170 m below the surface [64, 65]. A schematic overview of the LHC ring and its interaction points is shown in Fig. 4.1.

The LHC was proposed as the next major collider after LEP. The first circulating beams were achieved in 2008, and the first physics run began in 2010. The discovery of a new boson compatible with the Standard Model (SM) Higgs boson by the ATLAS and CMS experiments in 2012 marked one of the central achievements of the LHC physics programme [9, 10]. Beyond Higgs-boson measurements and precision studies of SM processes, a major purpose of the LHC is to search for new particles and interactions beyond the SM.

The LHC collides protons at high centre-of-mass energies in order to extend the kinematic reach for the production of heavy particles. For two proton beams with equal beam energy E_{beam} , the proton–proton centre-of-mass energy is approximately $\sqrt{s} = 2E_{\text{beam}}$. However, the hard-scattering process in a proton–proton (pp) collision occurs between partons, namely quarks and gluons, which carry only fractions of the proton momenta. The partonic centre-of-mass energy therefore varies from event to event and is generally lower than the nominal pp collision energy. Increasing $\sqrt{s}$ consequently extends the accessible mass range for new heavy particles.

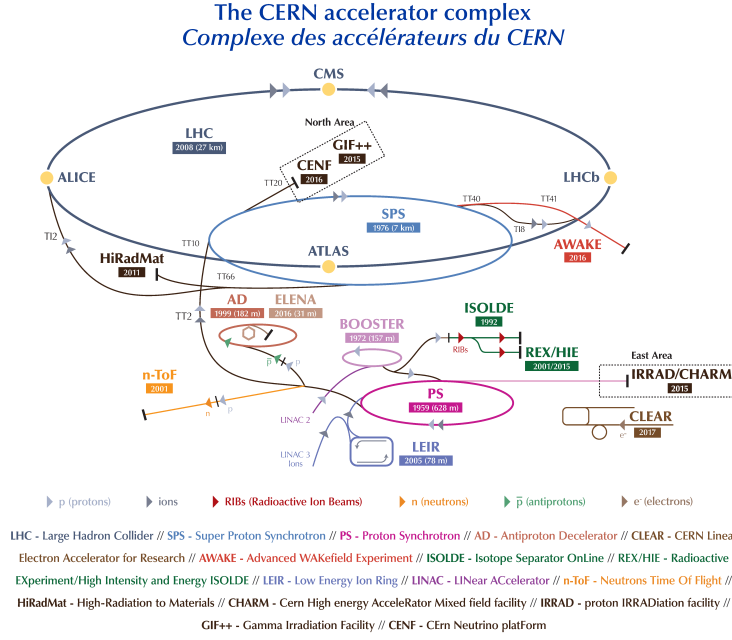


Fig. 4.1 Schematic layout of the CERN accelerator complex, including the LHC ring and its main interaction regions [66].

Another key machine parameter is the luminosity, which quantifies the collision rate per unit cross section. For a process with cross section σ , the production rate is

$$R = \mathcal{L}\sigma, \quad (4.1)$$

where $\mathcal{L}$ is the instantaneous luminosity. For two Gaussian bunches colliding with frequency f_{coll} , the luminosity can be expressed as [67]

$$\mathcal{L} = f_{\text{coll}} \frac{n_1 n_2}{4\pi\sigma_x^* \sigma_y^*} \mathcal{F}, \quad (4.2)$$

where n_1 and n_2 are the numbers of particles in the two colliding bunches, σ_x^* and σ_y^* are the RMS transverse beam sizes in the horizontal and vertical directions at the interaction point, and $\mathcal{F}$ is a geometrical reduction factor accounting for effects such as the crossing angle. The integrated luminosity,

$$\mathcal{L}_{\text{int}} = \int \mathcal{L} dt, \quad (4.3)$$

determines the total expected number of produced events by $N = \sigma \mathcal{L}_{\text{int}}$. High instantaneous luminosity increases the event rate, while high integrated luminosity improves the sensitivity to rare processes with small production cross sections.

Four major experiments are installed around the LHC ring. ATLAS, located at Point 1, and CMS, located at Point 5, are general-purpose detectors designed for a broad physics programme including Higgs-boson measurements, precision SM studies, and searches for new particles. ALICE, located at Point 2, is optimised for heavy-ion physics and the study of strongly interacting matter at high energy density. LHCb, located at Point 8, is designed for precision studies of heavy-flavour hadrons and flavour-changing processes [64]. The analysis presented in this thesis uses data recorded by the ATLAS detector.

The LHC physics programme is organised into run periods. Run 1 took place from 2010 to 2012, with pp collision data recorded at $\sqrt{s} = 7$ TeV and $\sqrt{s} = 8$ TeV. Run 2 took place from 2015 to 2018 at $\sqrt{s} = 13$ TeV. After standard data-quality selections, the full ATLAS Run 2 pp dataset corresponds to an integrated luminosity of $140.1 \pm 1.2 \text{ fb}^{-1}$, with a relative uncertainty of 0.83% [68]. Run 3 began in 2022 after the second long shutdown and operated at the higher centre-of-mass energy of $\sqrt{s} = 13.6$ TeV [69]. The Run 2 dataset is used for the leptoquark search presented in this thesis.

An important consequence of high-luminosity operation at the LHC is pile-up. The proton beams are divided into bunches, and a bunch crossing occurs when bunches from the two beams pass through the interaction point. Several pp interactions can occur in a single bunch crossing. The interaction corresponding to the hard-scattering process of interest is referred to as the hard-scatter interaction, while additional inelastic pp interactions in the same bunch crossing constitute in-time pile-up. Detector signals may also originate from interactions in neighbouring bunch crossings, depending on the detector response time. This contribution is referred to as out-of-time pile-up.

The mean number of inelastic pp collision per bunch crossing is commonly denoted by μ . During Run 2, the nominal bunch spacing was 25 ns, corresponding to a maximum bunch-crossing frequency of 40 MHz. Each of the bunch crossing over the full Run 2 period contains approximately $\langle \mu \rangle = 34$ [70].

Pile-up has an important impact on event reconstruction. It produce extra charged-particle tracks, calorimeter energy deposits, and reconstructed jets. They can affect the reconstruction of the primary vertex, jet energies, hadronically decaying τ -leptons, flavour-tagged jets, and missing transverse momentum. Pile-up effects must be therefore modelled in simulated samples and mitigated through dedicated reconstruction and calibration procedures.

4.2 ATLAS experiment

4.2.1 Overview

ATLAS, which stands for A Toroidal LHC ApparatuS, is a detector located at Point 1 of the Large Hadron Collider. It was designed to exploit the full physics potential of the LHC, including precision measurements of SM processes, studies of the Higgs boson, and searches for new physics beyond the SM. The detector surrounds the nominal interaction point and provides nearly full solid-angle coverage, allowing the reconstruction of a wide range of final-state particles produced in proton–proton collisions.

The ATLAS detector is approximately 44 m long, 25 m in diameter, and has a total weight of about 7000 tonnes [71]. It has a cylindrical, forward–backward symmetric layout centred on the interaction point. A schematic view of the detector is shown in Figure 4.2.

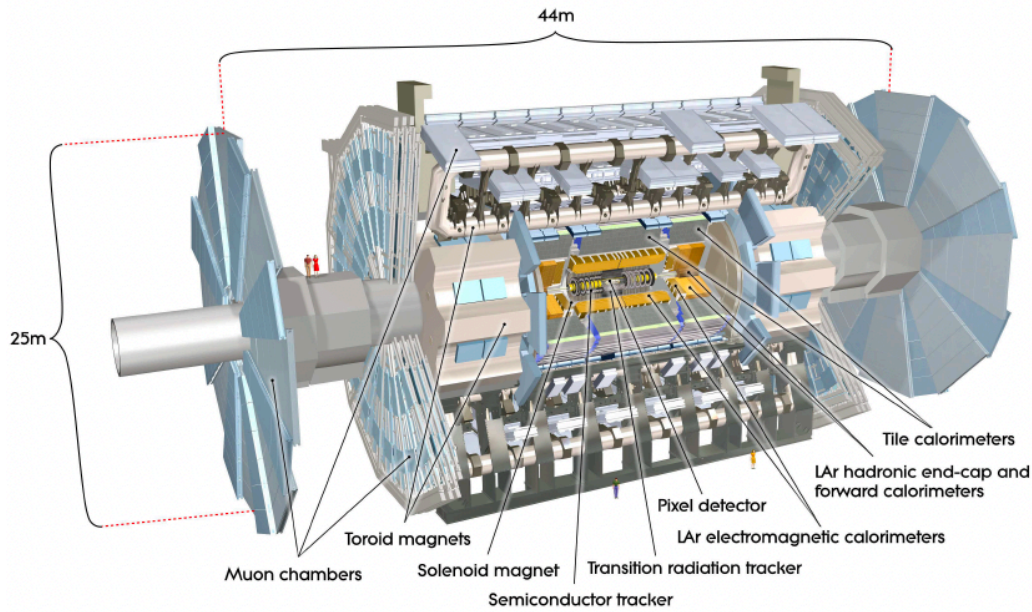


Fig. 4.2 Schematic view of the ATLAS detector and its subdetector systems [71].

Starting from the beam line and moving outwards, the main detector systems are the Inner Detector (ID), the calorimeters, and the Muon Spectrometer (MS) [71]:

- **Inner Detector (ID)** is immersed in a 2 T axial magnetic field provided by a thin superconducting solenoid and is used to reconstruct charged-particle trajectories close to the interaction point.
- **Calorimeters** surround the ID and measure the energies of electrons, photons, hadrons, hadronically decaying τ -leptons, and jets.

- **Muon Spectrometer (MS)** forms the outermost major detector system and measures muons in the magnetic field generated by large superconducting toroids.

The magnetic fields provided by the solenoid and toroids allow the momentum and charge of charged particles to be determined from the curvature of their trajectories. The trigger and data acquisition system selects potentially interesting events from the high LHC bunch-crossing rate for permanent storage and offline reconstruction.

The detector configuration described in this chapter corresponds primarily to Run 2. Several detector and trigger upgrades were installed for Run 3 to cope with higher luminosity conditions, but these upgrades are not used in the analysis presented in this thesis [72].

4.2.2 Coordinate system

ATLAS uses a right-handed coordinate system centred at the nominal interaction point. The z -axis is defined along the beam direction, the x -axis points from the interaction point towards the centre of the LHC ring, and the y -axis points upwards [71]. Cylindrical coordinates (r, ϕ) are used in the plane transverse to the beam, where r is the distance from the beam axis and ϕ is the azimuthal angle around the z -axis.

The polar angle θ is measured with respect to the positive z -axis. The pseudorapidity η is defined as

$$\eta = -\ln \tan \left(\frac{\theta}{2} \right). \quad (4.4)$$

The transverse momentum is defined as $p_T = p \sin \theta$, and the transverse energy is defined similarly. The angular separation between two reconstructed objects is commonly expressed as

$$\Delta R = \sqrt{(\Delta \eta)^2 + (\Delta \phi)^2}. \quad (4.5)$$

These quantities are used throughout this thesis to describe object kinematics, angular separations, overlap removal, and event selections. Figure 4.3 illustrates the coordinate system used in ATLAS.

4.2.3 Magnet system

The ATLAS magnet system consists of four large superconducting magnets: a central solenoid, one barrel toroid, and two end-cap toroids, as shown in Figure 4.4. The central solenoid surrounds the Inner Detector with a length of approximately 5.3 m and provides an

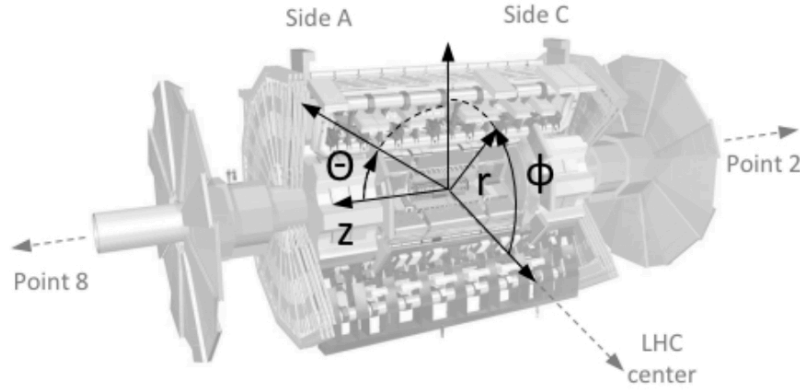


Fig. 4.3 The coordinate system used in ATLAS [73].

axial magnetic field of 2 T for charged-particle momentum measurements in the tracking system. The solenoid is designed to minimise the amount of inactive material in front of the electromagnetic calorimeter.

The toroid system provides magnetic fields of approximately 0.5 T and 1 T in the barrel and end-cap regions of the Muon Spectrometer, respectively. The barrel toroid surrounds the calorimeter system in the central region, while the two end-cap toroids provide bending power in the forward regions. The toroidal field bends muon trajectories, allowing their momenta to be measured using the precision tracking chambers of the MS. The air-core design of the toroids reduces multiple scattering and helps preserve the momentum resolution for high- p_T muons [71]. The combination of the solenoid and toroid systems allows charged-particle momenta to be measured independently in the ID and, for muons, in the MS.

4.2.4 Inner Detector

The Inner Detector (ID) is the innermost tracking system of ATLAS. It is immersed in the 2 T magnetic field of the central solenoid and provides tracking for charged particles within $|\eta| < 2.5$ [71]. During Run 2, it consisted of the pixel detector, the Semiconductor Tracker (SCT), and the Transition Radiation Tracker (TRT). The Insertable B-Layer (IBL) is installed before Run 2 as the innermost pixel layer [74]. Figure 4.5 shows a transverse view of the Inner Detector.

- ***Insertable B-Layer (IBL) and Pixel detector***

The pixel detector provides high-granularity measurements close to the beam line. Its precise spatial resolution is essential for reconstructing tracks near the interaction point and for identifying displaced vertices from the decays of long-lived particles such

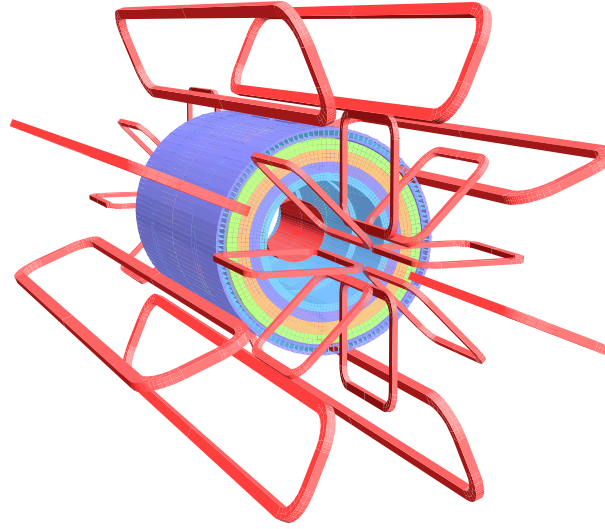


Fig. 4.4 Layout of the ATLAS superconducting magnet system, including the central solenoid, barrel toroid, and end-cap toroids [71].

as b - and c -hadrons. The IBL further improves the impact-parameter resolution and enhances the robustness of tracking and flavour tagging under high pile-up conditions.

- ***Semiconductor Tracker (SCT)***

The SCT surrounds the pixel detector and forms the intermediate precision-tracking system of the ID. It is a silicon microstrip detector covering $|\eta| < 2.5$, consisting of four cylindrical barrel layers and two end-caps with nine disks on each side. Each module contains silicon strip sensors arranged back-to-back with a small stereo angle, allowing two-dimensional position information to be reconstructed. The SCT typically provides four space-point measurements per track, improving pattern recognition, momentum reconstruction, and vertex reconstruction when combined with the high-granularity pixel measurements.

- ***Transition Radiation Tracker (TRT)***

The TRT is the outermost tracking subsystem of the ID and provides tracking information at larger radii, covering approximately $|\eta| < 2.0$ [71]. It is based on thin straw-tube drift chambers, which provide a large number of measurements along the trajectory of each charged particle. Although the spatial precision of each TRT hit is lower than that of the silicon detectors, the large number of straw hits and the longer lever arm improve the momentum measurement and help stabilise track reconstruction.

Together, these subsystems provide the tracking information needed in ATLAS.

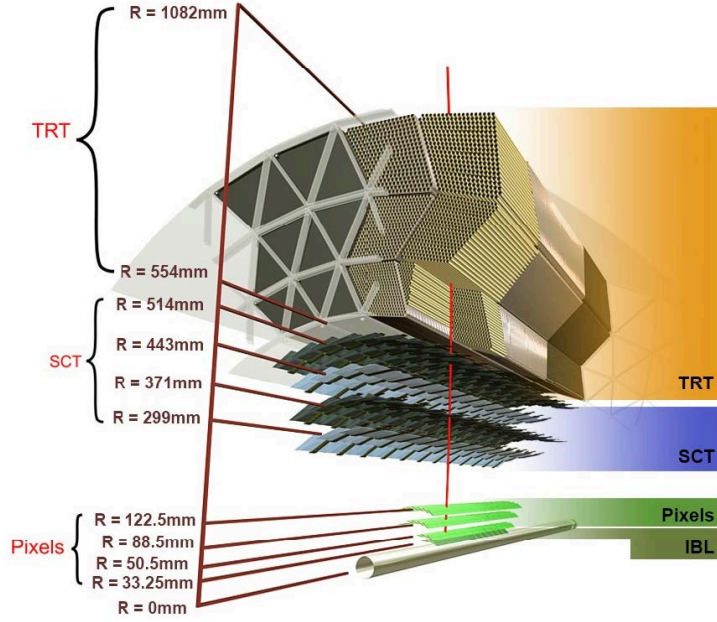


Fig. 4.5 Transverse view of the ATLAS Inner Detector, showing the IBL, pixel detector, SCT, and TRT subsystems [74].

4.2.5 Calorimeter

The ATLAS calorimeter system surrounds the Inner Detector and measures the energies and directions of particles. It consists of electromagnetic (EM) and hadronic calorimeters, as shown in Figure 4.6. Specifically, it provides measurements of charged particles, photons and jets, and covers the region up to $|\eta| < 4.9$. It also provides the shower information to the Level-1 trigger system.

- ***Electromagnetic (EM) calorimeter***

The EM calorimeter uses liquid argon (LAr) as the active medium and lead absorbers arranged in an accordion geometry, as shown in Figure 4.7. The accordion geometry provides continuous azimuthal coverage and fine segmentation of electromagnetic showers [71]. The EM calorimeter is divided into a barrel section covering $|\eta| < 1.475$ and two end-cap sections covering $1.375 < |\eta| < 3.2$. It is used to reconstruct and identify electrons and photons and also contributes to the reconstruction of hadronically decaying τ -leptons, particularly through the measurement of electromagnetic energy deposits from neutral-pion decays, which don't leave the charged-particle tracks in the ID.

The EM calorimeter consists of a presampler followed by three longitudinal sampling layers, each segmented into readout cells, as illustrated in Figure 4.8. The presampler

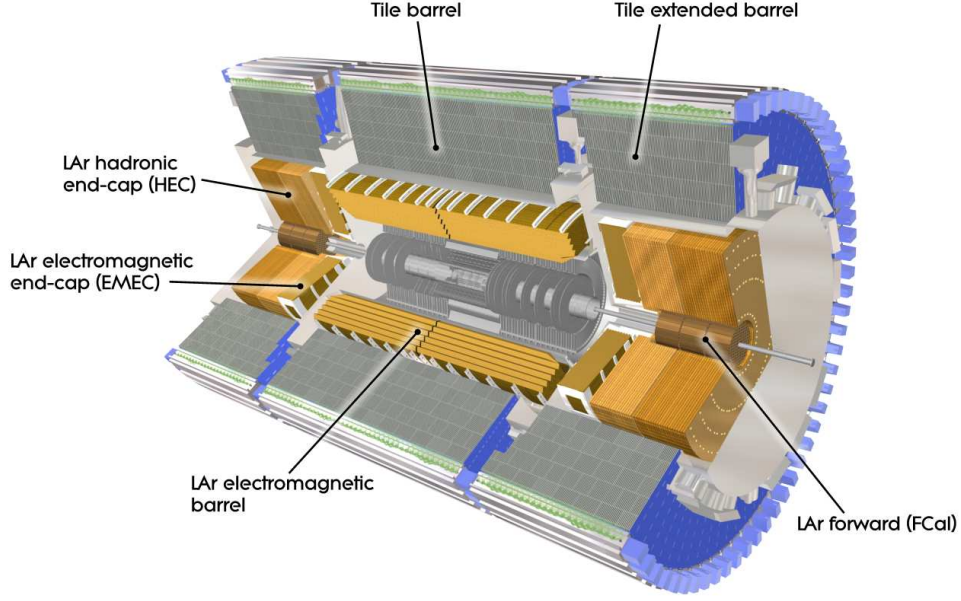


Fig. 4.6 Schematic view of the ATLAS calorimeter system [71].

is installed for $|\eta| < 1.8$ and is used to correct for energy lost in material upstream of the calorimeter. The first sampling layer has fine segmentation in η , which is useful for resolving the transverse shape of electromagnetic showers, while the middle layer collects the largest fraction of the shower energy. The back layer provides an additional measurement of the longitudinal shower development.

The energy resolution of the electromagnetic calorimeter can be parameterised as

$$\frac{\sigma_E}{E} = \frac{a}{\sqrt{E}} \oplus b \oplus \frac{c}{E}, \quad (4.6)$$

where a is the intrinsic sampling term, b is the constant term, and c represents the contribution from electronic and pile-up noise [75].

- **Hadronic calorimeter**

The hadronic calorimeter uses different structure for different η regions. In the region of $|\eta| < 1.7$, the Tile Calorimeter uses steel absorbers and scintillating tiles as the active medium. It measures the energies of hadrons and jets. In the end-cap and forward regions, liquid-argon calorimeters are used. The hadronic end-cap calorimeter consists of two wheels per end-cap, covers the range $1.5 < |\eta| < 3.2$, and uses copper absorbers. The forward calorimeter extends the coverage to $3.1 < |\eta| < 4.9$ and uses

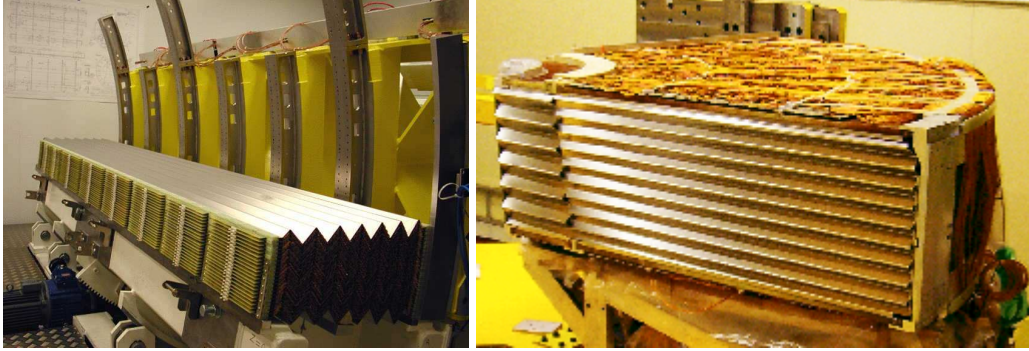


Fig. 4.7 Photograph of the EM barrel (left) and end-cap (right) LAr modules. The accordion absorber structure is visible [71].

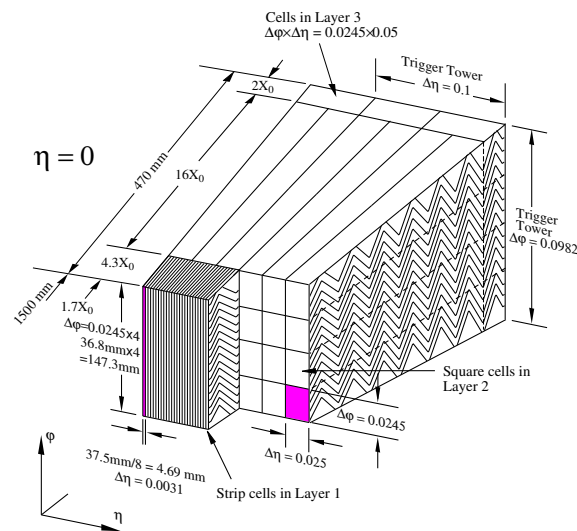


Fig. 4.8 Cut-away view of an ATLAS LAr electromagnetic calorimeter module, showing the longitudinal sampling layers and the segmentation of the readout cells [71].

dense absorber materials suitable for the high-radiation environment close to the beam line [71].

4.2.6 Muon Spectrometer

The Muon Spectrometer (MS) forms the outermost major detector system of ATLAS. It is embedded in the magnetic field produced by the barrel and end-cap air-core toroid magnets. The bending of muon trajectories in this field is measured by precision chambers, and the track sagitta is used to determine the muon momentum. The MS provides precision muon measurements in the range $|\eta| < 2.7$, while the dedicated muon trigger chambers cover the region up to approximately $|\eta| < 2.4$ [71].

During Run 2, the MS consisted of Monitored Drift Tubes (MDTs), Cathode Strip Chambers (CSCs), Resistive Plate Chambers (RPCs) and Thin Gap Chambers (TGCs) [71]. A view of these chambers is shown in Figure 4.9.

- ***Monitored Drift Tubes (MDTs)***

The MDT chambers provide precision tracking over most of the MS acceptance. They consist of aluminium drift tubes with a diameter of about 30 mm, each containing a central sense wire. A muon crossing a tube ionises the gas, and the drift time of the electrons to the wire is measured to determine the track position. Several tube layers are grouped into chamber stations, providing precise measurements in the bending plane of the toroidal magnetic field. MDT chambers are used over most of the region $|\eta| < 2.7$, except for the innermost forward stations at $2.0 < |\eta| < 2.7$, where CSCs are used.

- ***Cathode Strip Chambers (CSCs)***

The CSCs are multi-wire proportional chambers used in the innermost forward muon stations, where the particle rates are highest. They cover approximately $2.0 < |\eta| < 2.7$ and provide precision measurements in the forward region [71]. Their small drift distance and segmented cathode-strip readout allow them to operate efficiently in the high-rate environment close to the beam line.

- ***Resistive Plate Chambers (RPCs)***

The RPCs provide fast muon trigger information in the barrel region, covering approximately $|\eta| < 1.05$ [71]. They are gaseous parallel-plate detectors operated in avalanche mode. Their fast time response allows the bunch crossing associated with a muon candidate to be identified and provides the Level-1 muon trigger in the barrel.

The RPCs also provide coordinate measurements complementary to the precision MDT chambers.

- **Thin Gap Chambers (TGCs)**

The TGCs are multi-wire proportional chambers designed to provide fast signals for the Level-1 trigger. They extend to approximately $|\eta| < 2.7$ and provide trigger coverage in the end-cap region up to $|\eta| < 2.4$ [71]. The TGC system identifies high- p_T muon candidates and assigns them to the correct bunch crossing. It also provides a second coordinate measurement, complementing the precision tracking information from MDTs and CSCs.

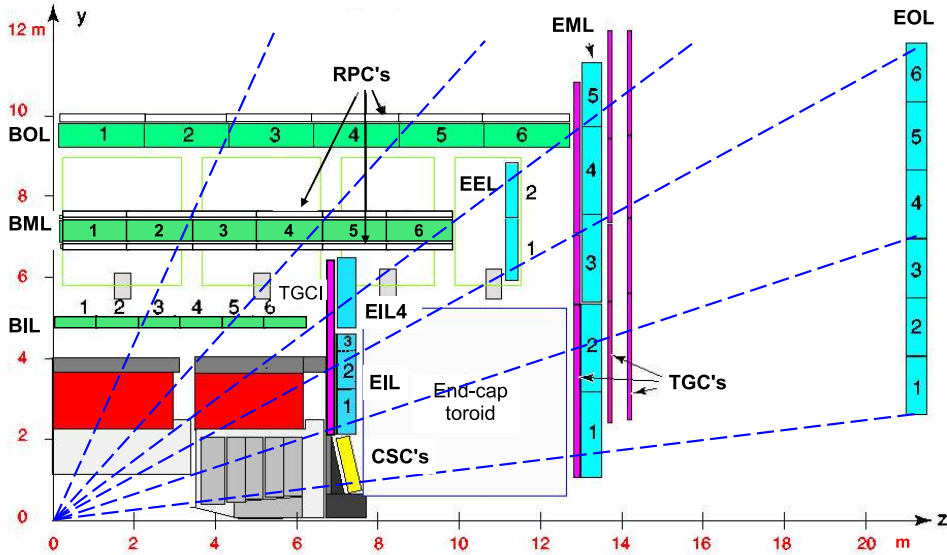


Fig. 4.9 Cross-section of the ATLAS Muon Spectrometer, showing the barrel and end-cap muon chambers. The green and cyan chambers represent the MDT system, while the dashed lines indicate infinite-momentum muon trajectories [71].

During the second long shutdown, the innermost end-cap muon stations were replaced by the New Small Wheels (NSWs) for Run 3 operation. This upgrade was introduced to maintain good muon trigger and reconstruction performance under higher luminosity and pile-up conditions. The NSW detectors are therefore not relevant to the Run 2 dataset analysed in this thesis.

4.2.7 Data Acquisition and Trigger System

At the LHC, proton bunches can cross at a rate of up to 40 MHz for a bunch spacing of 25 ns. This rate is far above the available detector readout and permanent-storage bandwidth. The ATLAS trigger and data acquisition (TDAQ) system is therefore designed to select potentially interesting events online and reduce the event rate to a manageable level. ATLAS uses a two-level trigger system consisting of a hardware-based Level-1 (L1) trigger and a software-based High-Level Trigger (HLT) [76].

Level-1 (L1) trigger

The L1 trigger is the first stage of the ATLAS online event selection. It is implemented in custom hardware and uses reduced-granularity detector information to make a fast decision with a latency of approximately $2.5 \mu\text{s}$. It reduces the input rate from up to 40 MHz to at most 100 kHz [76]. The L1 trigger was based on information from the calorimeter and muon trigger systems, together with topological selections applied to L1 trigger objects. A schematic overview of the Run 2 TDAQ system is shown in Figure 4.10.

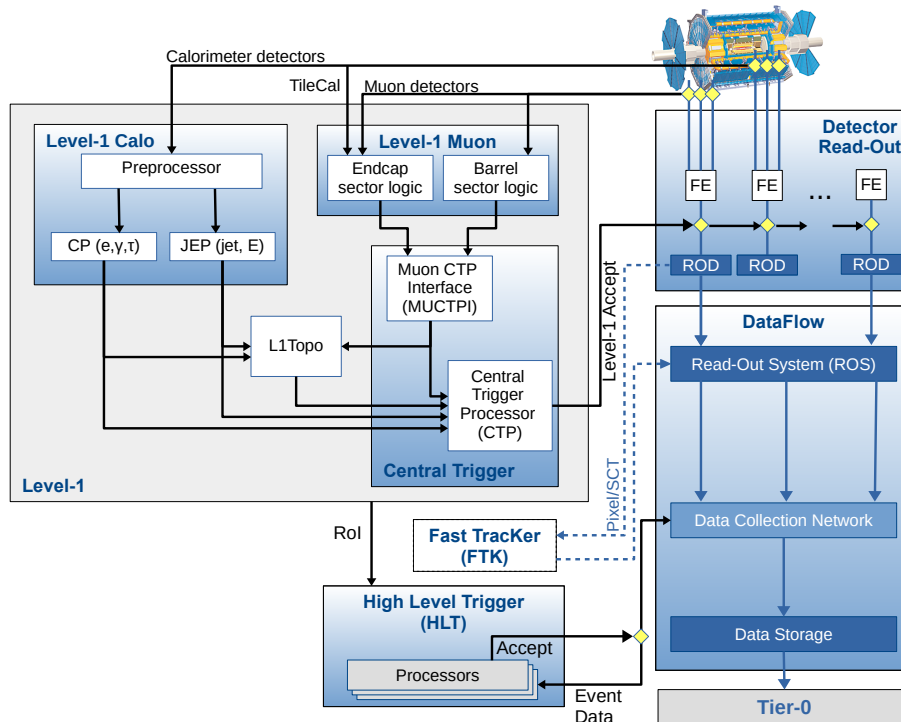


Fig. 4.10 Schematic overview of the ATLAS Run-2 trigger and data acquisition system [76].

- ***L1 calorimeter trigger (L1Calo)***

L1Calo uses reduced-granularity information from the electromagnetic and hadronic calorimeters to reconstruct energy deposits within the Level-1 latency. Signals from neighbouring calorimeter cells are combined into trigger towers in the (η, ϕ) plane, and dedicated hardware algorithms search for local maxima in the transverse-energy distribution by comparing the energies of neighbouring towers. The same calorimeter information is also summed over larger regions to reconstruct jet-like energy deposits and global quantities such as the total and missing transverse energy.

- ***L1 muon trigger (L1Muon)***

L1Muon identifies muon candidates using fast signals from the RPCs and the TGCs. Hits in different trigger-chamber layers are required to form spatial coincidence patterns consistent with a muon trajectory, while their timing information is used to associate the candidate with the correct bunch crossing. The amount of bending in the magnetic field is used to determine whether the candidate satisfies predefined transverse-momentum thresholds.

- ***L1 topological trigger (L1Topo)***

L1Topo receives compact trigger-object information reconstructed by L1Calo and L1Muon, including their transverse momentum or energy and (η, ϕ) positions. Dedicated FPGA hardware combines these objects and calculates event-level topological quantities. And these quantities are compared with predefined selection requirements.

The trigger decisions from L1Calo, L1Muon, and L1Topo are then sent to the Central Trigger Processor (CTP), which combines them according to the predefined L1 trigger menu.

If at least one enabled trigger item is satisfied, the CTP issues a Level-1 Accept (L1A) signal for the corresponding bunch crossing. The L1A signal is distributed to the detector front-end electronics and initiates the transfer of the detector data associated with that bunch crossing from the front-end buffers to the readout system.

For accepted events, the L1 trigger also provides Regions of Interest (RoIs), which identify detector regions associated with the trigger objects responsible for the L1 decision. These RoIs provide seed information for the HLT and allow detailed reconstruction to be performed initially in limited detector regions, reducing the required processing time.

High-Level Trigger

The HLT is the second stage of the ATLAS trigger system. It is implemented in a large computing farm and runs software algorithms using detector information with substantially

finer granularity than is available at L1. The HLT reconstruction is guided by RoIs identified by the L1 trigger, while full-event reconstruction can also be used for signatures such as jets and missing transverse momentum. The HLT algorithms are designed to provide reconstruction performance close to that of the offline algorithms while remaining sufficiently fast for online operation. As a result, the HLT reduced the L1 output rate to an average recording rate of approximately 1 kHz for permanent storage [76].

The trigger menu comprises the active L1 and HLT selections used during data taking with prescale factors, which are applied to selected trigger items to control their rates according to the instantaneous luminosity and available bandwidth.